

Two dimensional array:

Two dimensional arrays:

Array with several rows and columns.

Array with two subscripting operators **[] []**.

It is array of arrays. i.e. collection of one-dimensional arrays.

It is implicit double pointer.

It is a $n \times n$ matrix.

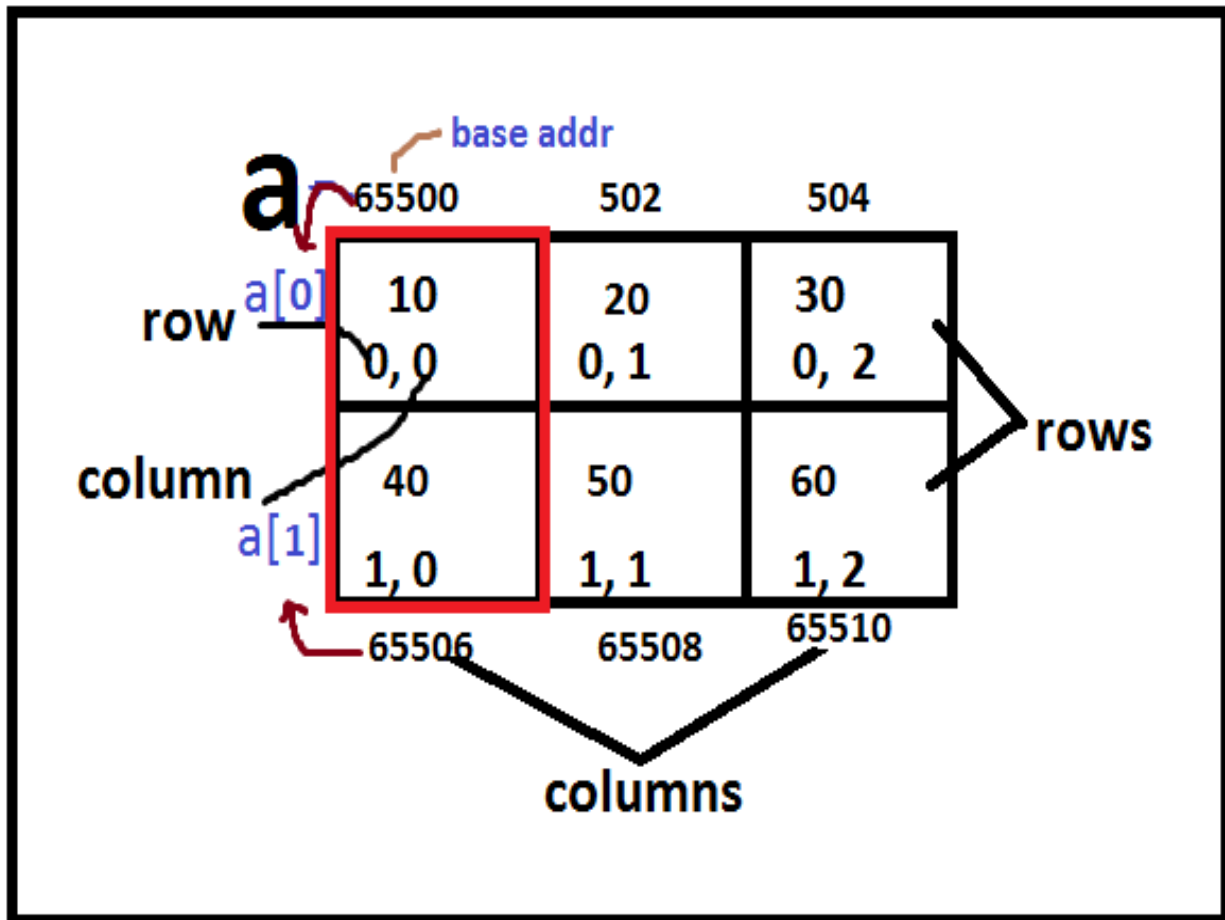
Syntax:

datatype variable [rows] [columns] = {elements} ;

Eg:

```
int a[ 2 ][ 3 ] = { { 10, 20 , 30 } , { 40 ,50, 60 } };
```

stack



In two dimensional array the rows/first subscript is working as array of pointers and they stores first column address of each row. Hence it is an implicit/internal double pointer.

In the above example, To print the first row, first column value, we have to use

`printf("%d", a[0][0]);` → 10

Internally how this statement is working ?

`a[0]` means value at `a[0]` i.e. 65500.

`65500 + [0] col` → `65500 + 0 * 2` → `65500` →
value at 65500 is 10.

Index no

Int size

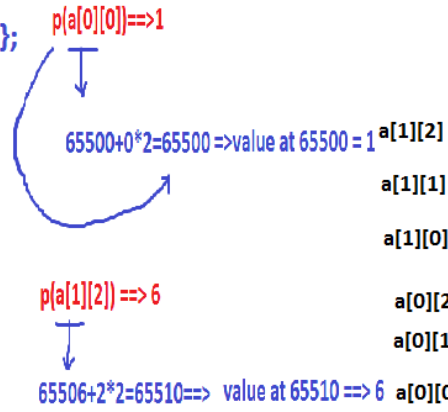
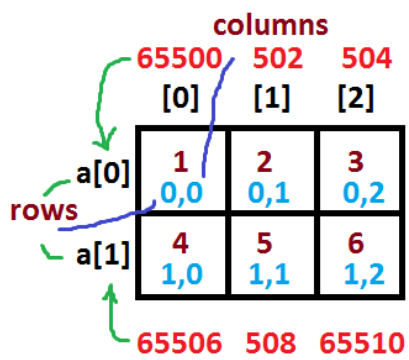
stack

<code>a[1][2]</code>	<u>60</u>	65500
<code>a[1][1]</code>	<u>50</u>	65498
<code>a[1][0]</code>	40	65496
<code>a[0][2]</code>	<u>30</u>	65494
<code>a[0][1]</code>	<u>20</u>	65492
<code>a[0][0]</code>	<u>10</u>	65490

`int a[2][2]={10,20,30,40};`

<code>a[1][1]</code>	40	65496
<code>a[1][0]</code>	<u>30</u>	65494
<code>a[0][1]</code>	<u>20</u>	65492
<code>a[0][0]</code>	<u>10</u>	65490

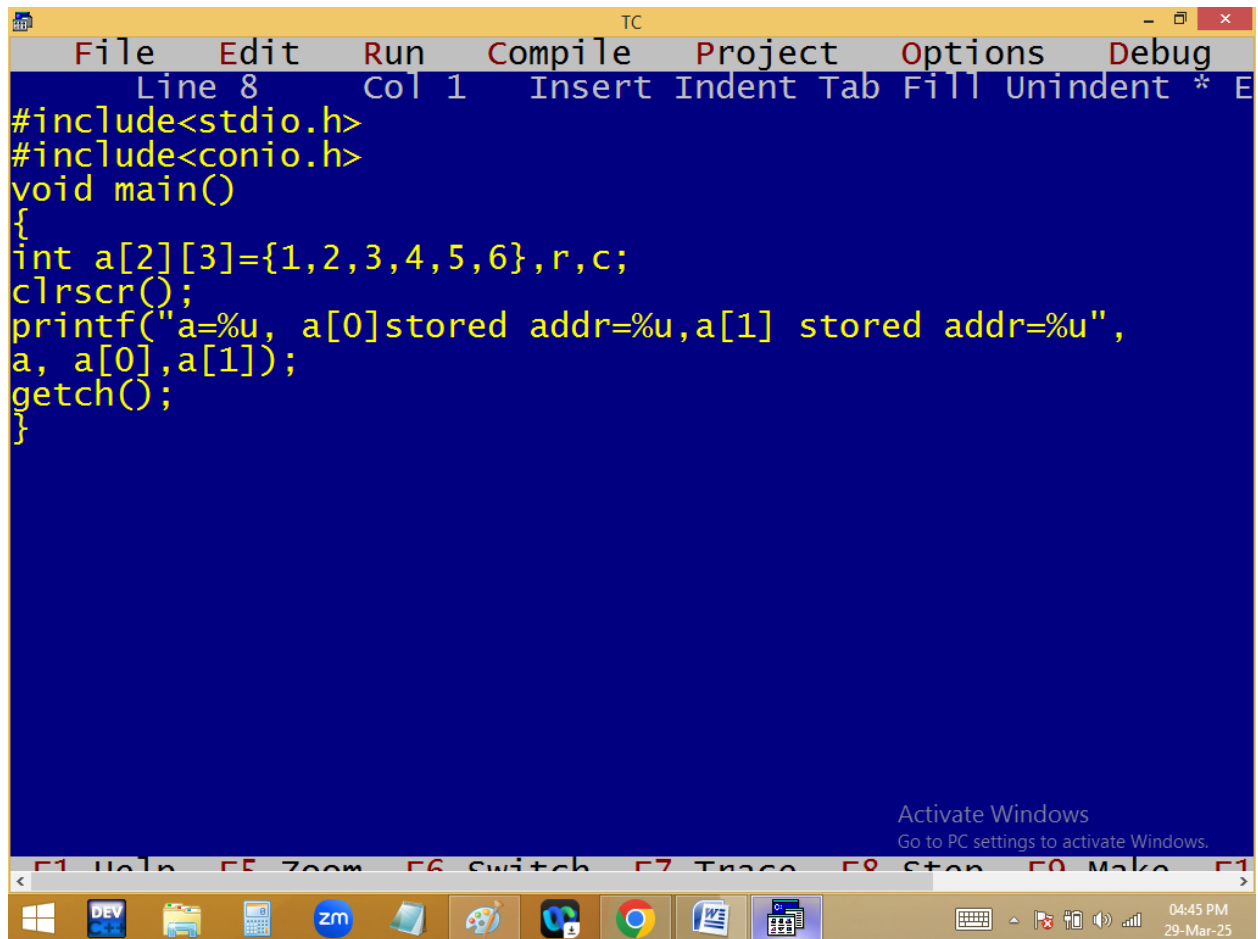
```
int a[2][3] = { { 1,2,3 }, { 4,5,6 } };
```



stack

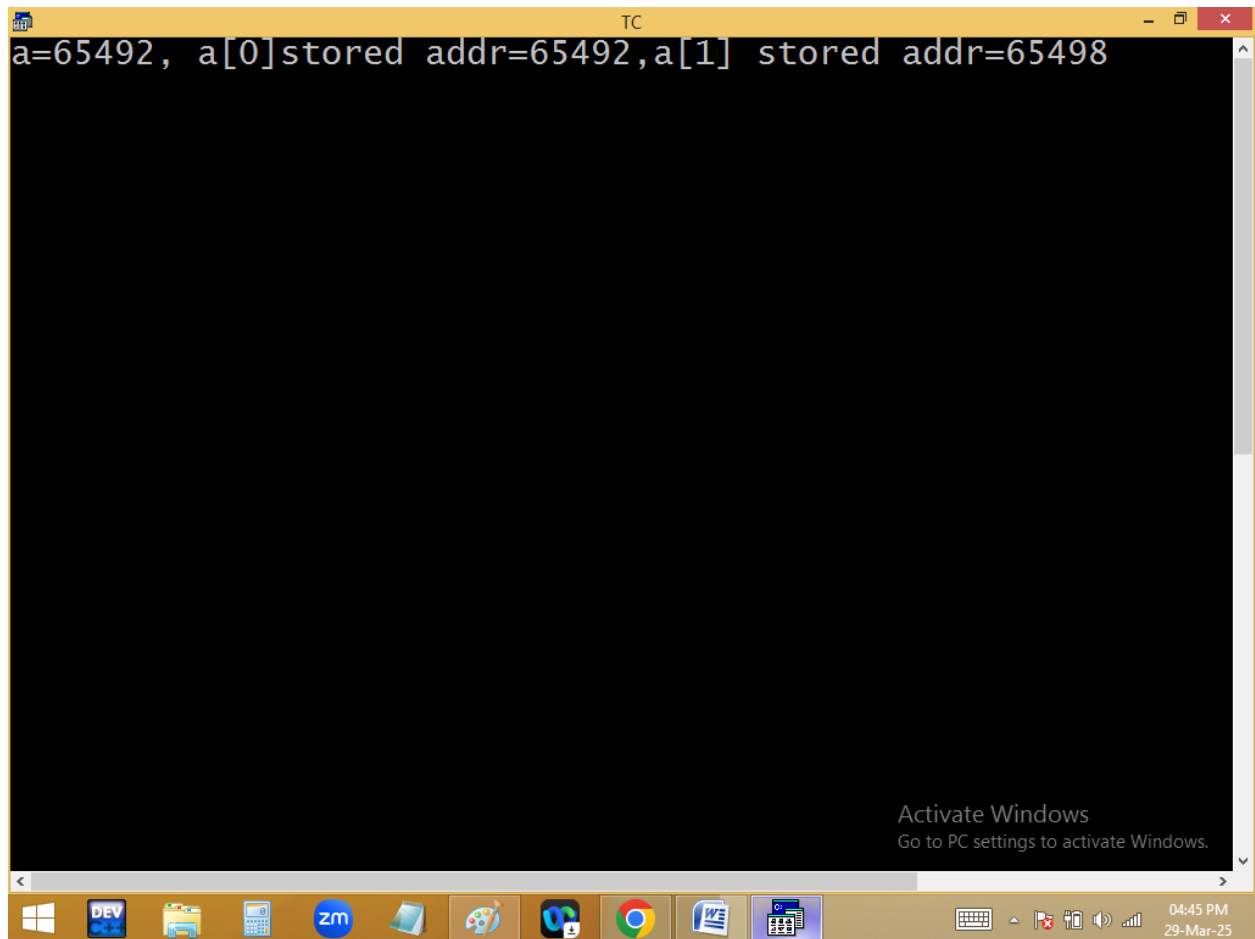
<code>a[1][2]</code>	6	65510
<code>a[1][1]</code>	5	65508
<code>a[1][0]</code>	4	65506
<code>a[0][2]</code>	3	65504
<code>a[0][1]</code>	2	65502
<code>a[0][0]</code>	1	65500

Finding row address:



```
TC
File Edit Run Compile Project Options Debug
Line 8 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
printf("a=%u, a[0]stored addr=%u,a[1] stored addr=%u",
a, a[0],a[1]);
getch();
}

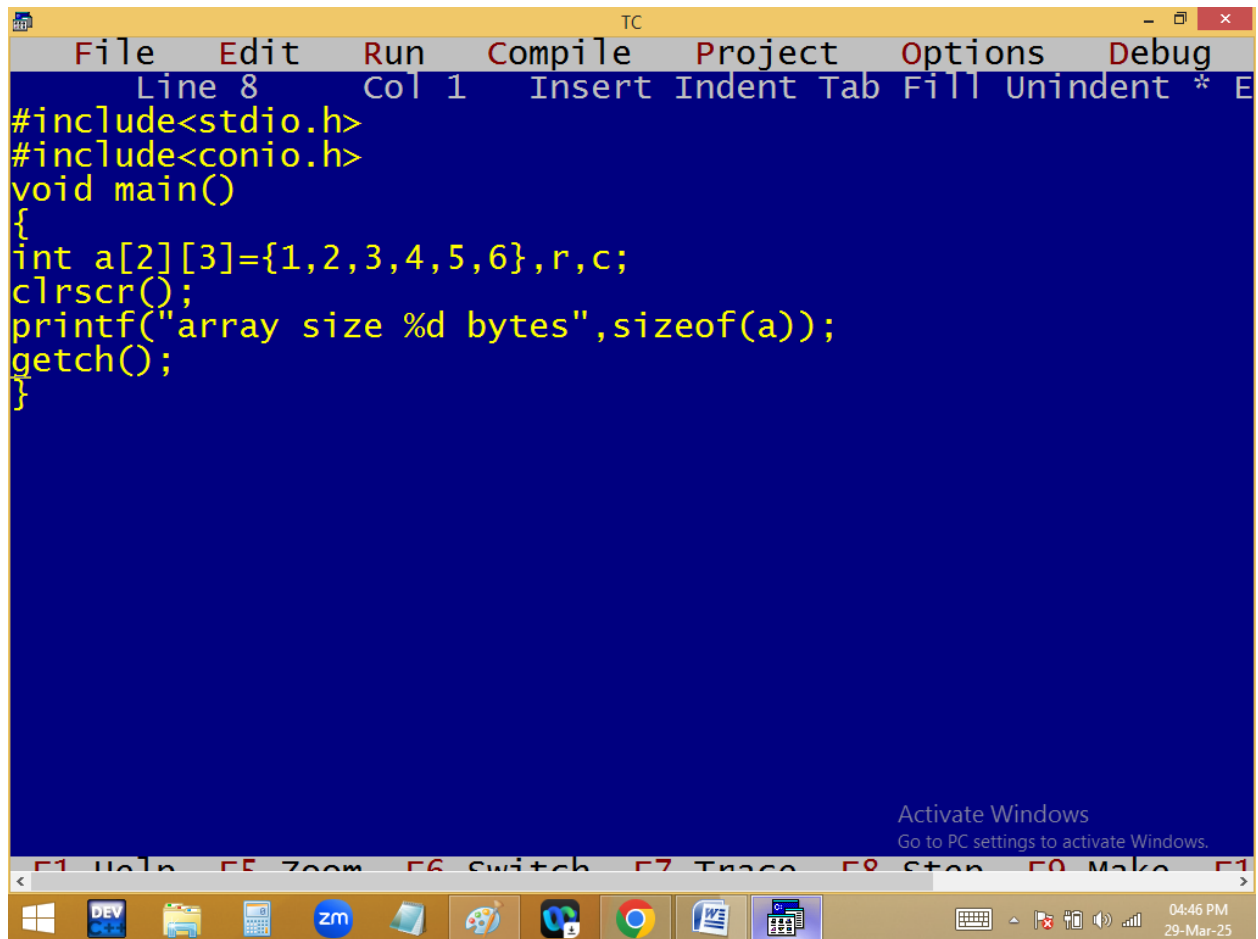
Activate Windows
Go to PC settings to activate Windows.
F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



The image shows a screenshot of a Turbo C++ (TC) compiler window. The title bar of the window is yellow and contains the text "TC". The main area of the window is black and displays the following text in white: `a=65492, a[0] stored addr=65492, a[1] stored addr=65498`. In the bottom right corner of the black area, there is a white text prompt: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom of the screen, showing various application icons including Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right side of the taskbar shows the time as 04:45 PM and the date as 29-Mar-25.

Finding array size:

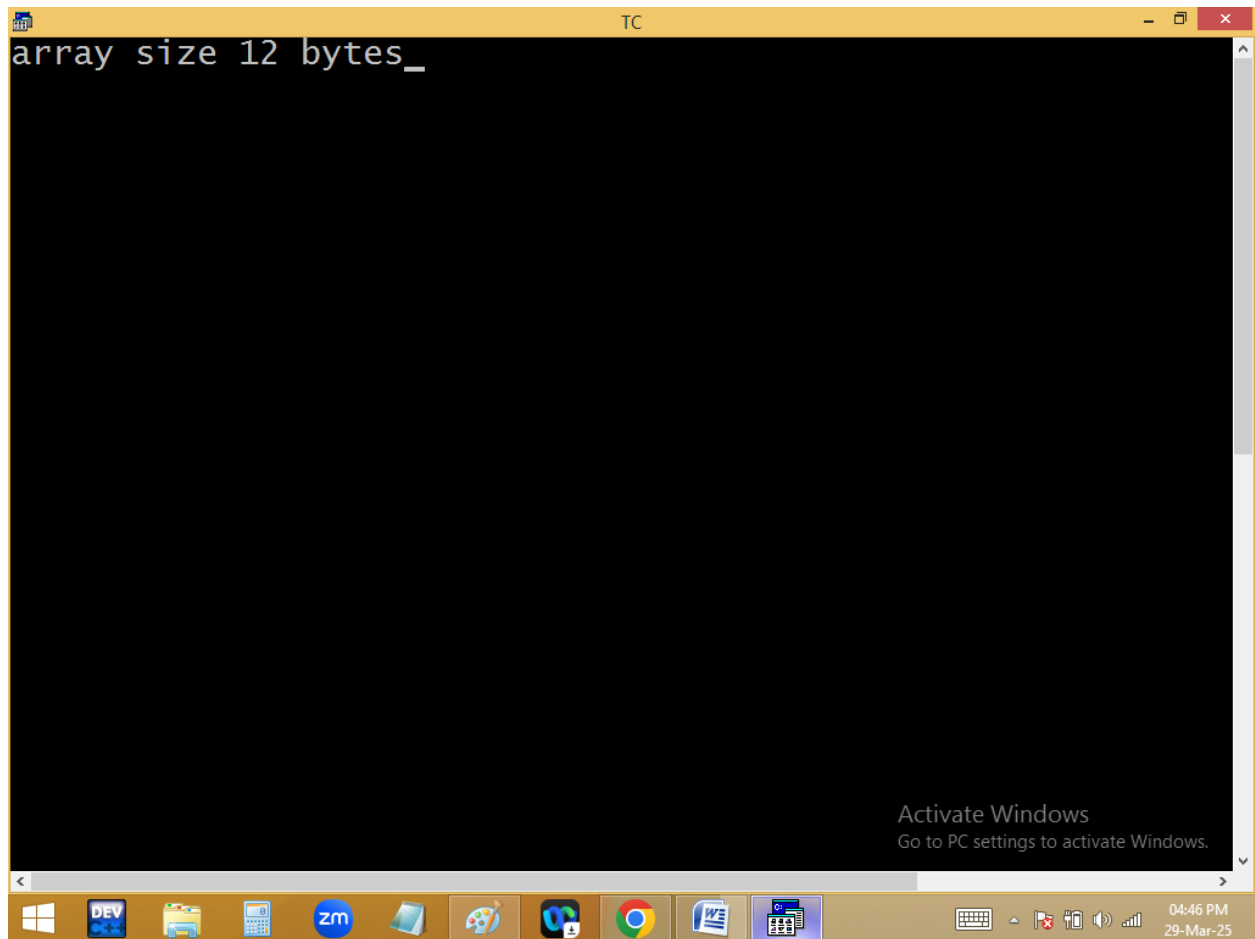
In C & C++:



The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 8" and "Col 1". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
printf("array size %d bytes",sizeof(a));
getch();
}
```

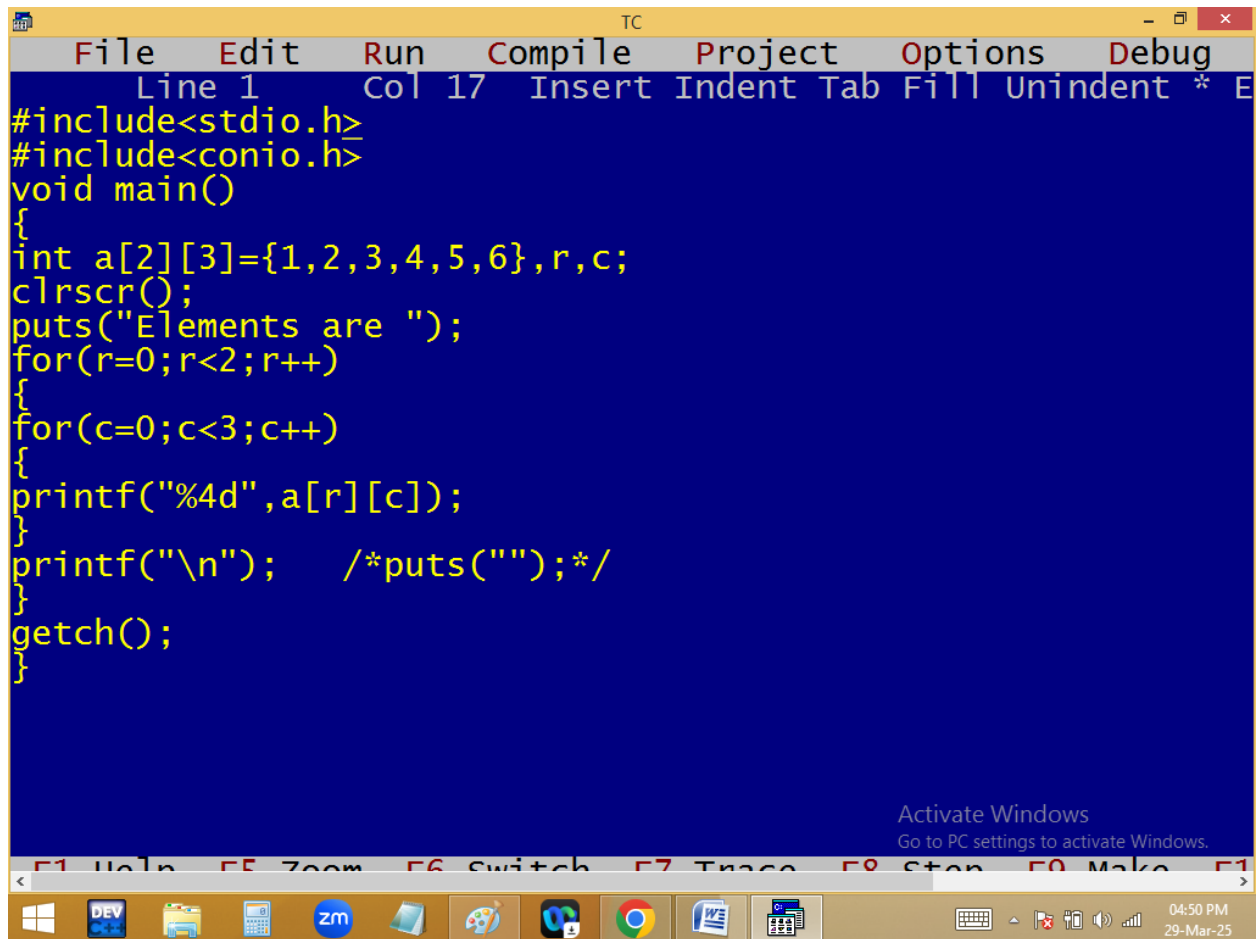
At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time as "04:46 PM 29-Mar-25".



In VSCode / Dev C++ / java / .net / online compilers:

24 bytes [any 32 bit compilers]

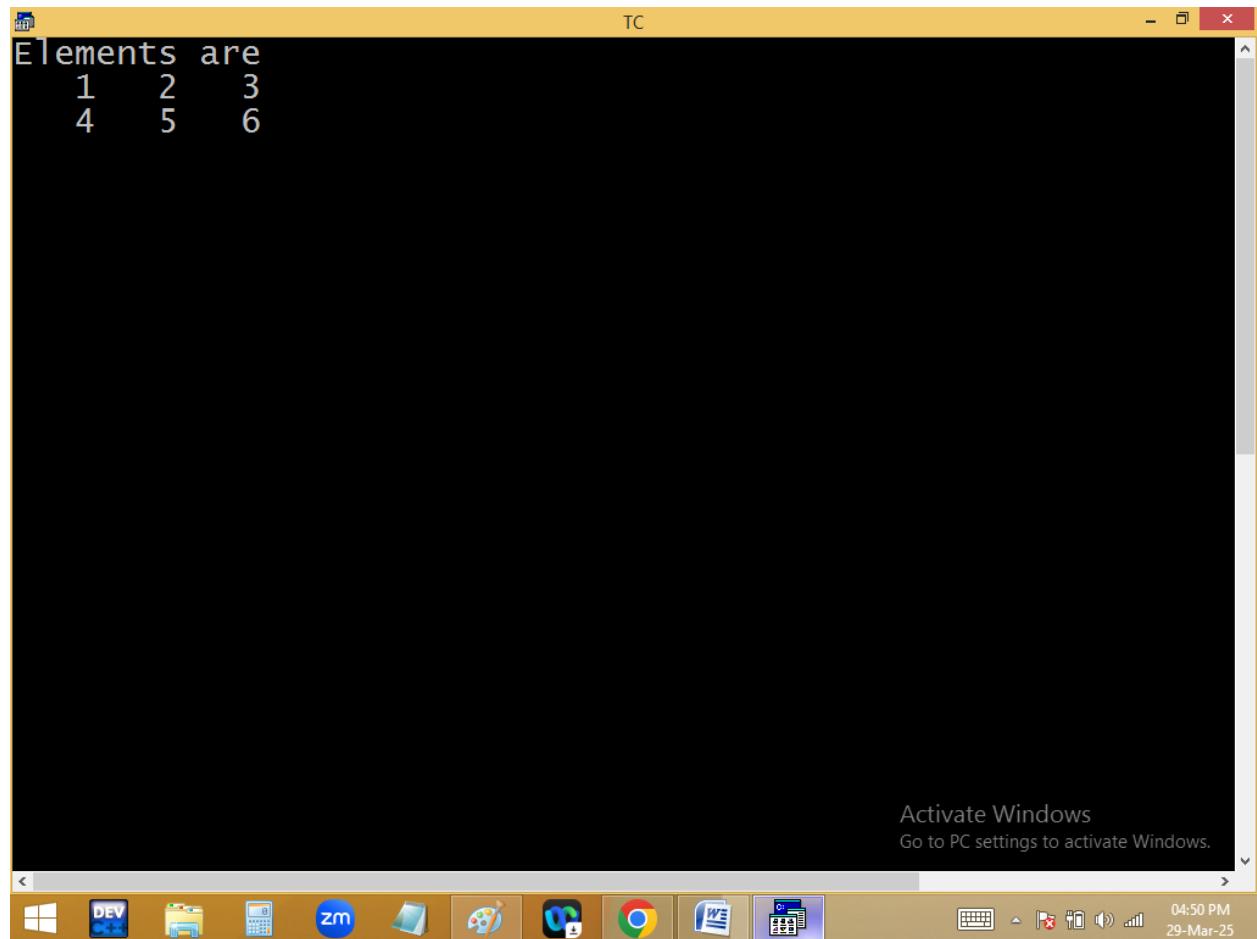
Direct initialization of 2*3 matrix elements:

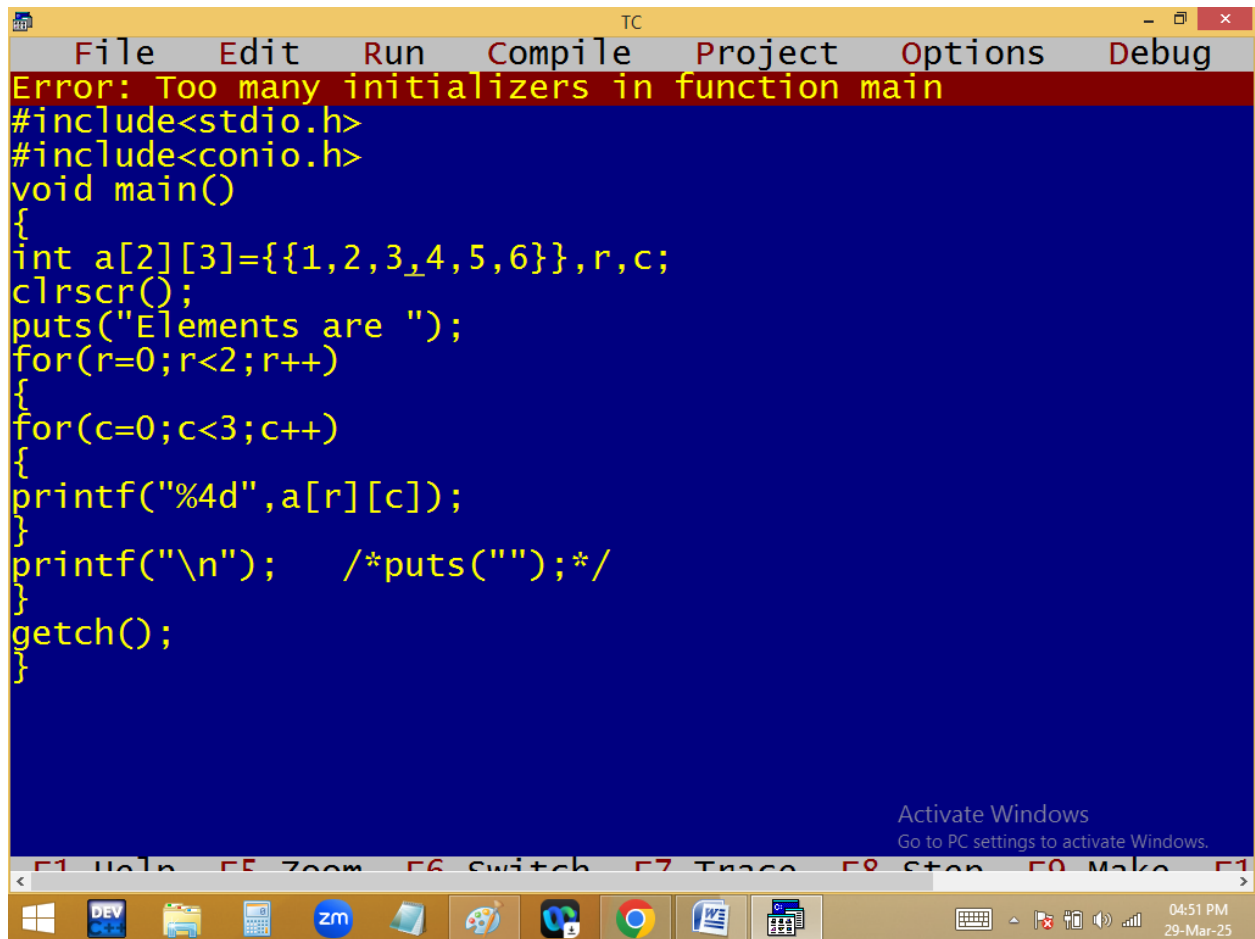


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 1", "Col 17", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a taskbar with icons for Windows, DEV, File Explorer, Calculator, Zoom, and other applications. The system tray on the right shows the time as "04:50 PM" and the date as "29-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code editor displays the message: "Error: Too many initializers in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3,4,5,6}},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons (Windows logo, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, Excel) and a system tray showing the time as 04:51 PM on 29-Mar-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

TC

File Edit Run Compile Project Options Debug

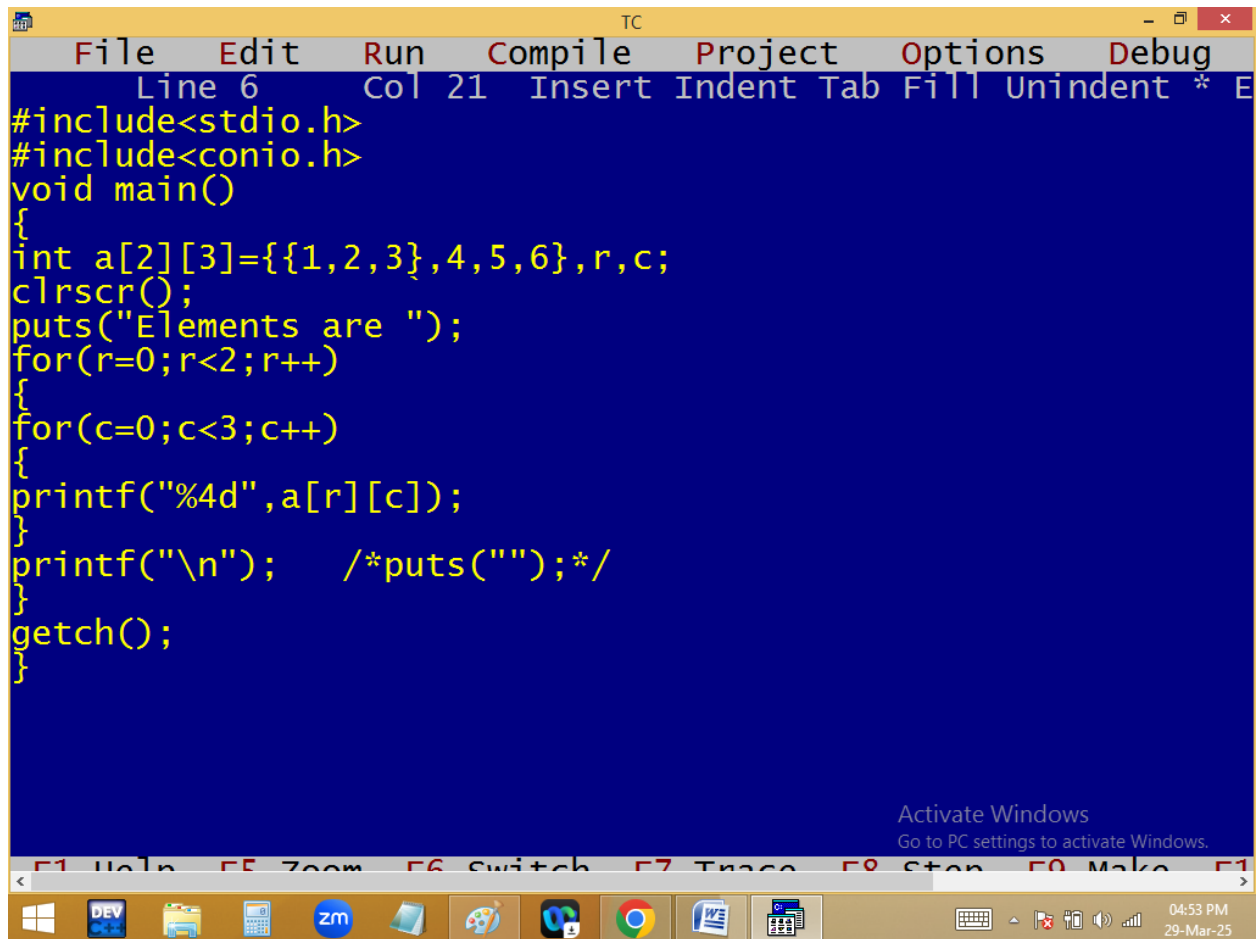
Error: Too many initializers in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3,4},5,6},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

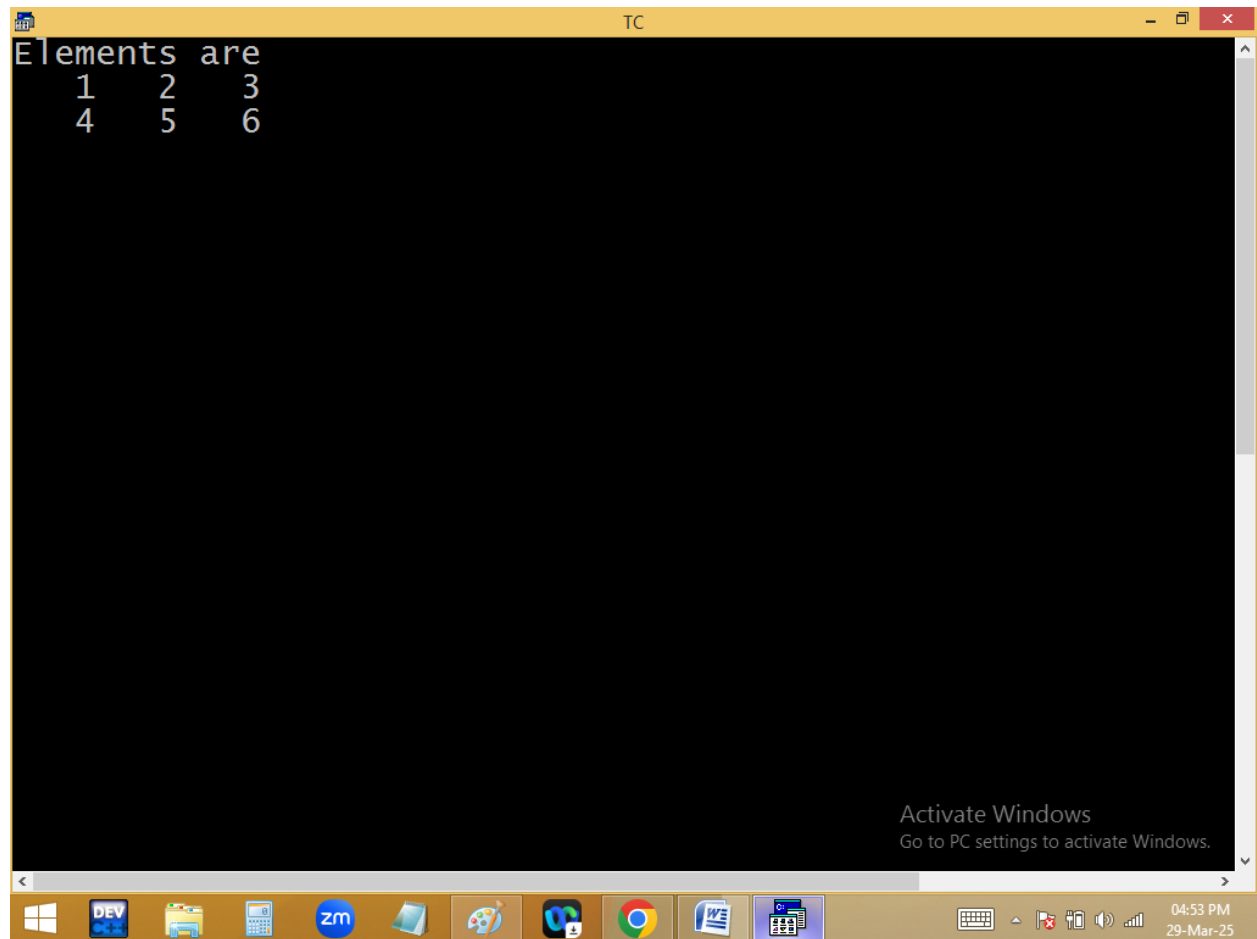
04:52 PM
29-Mar-25

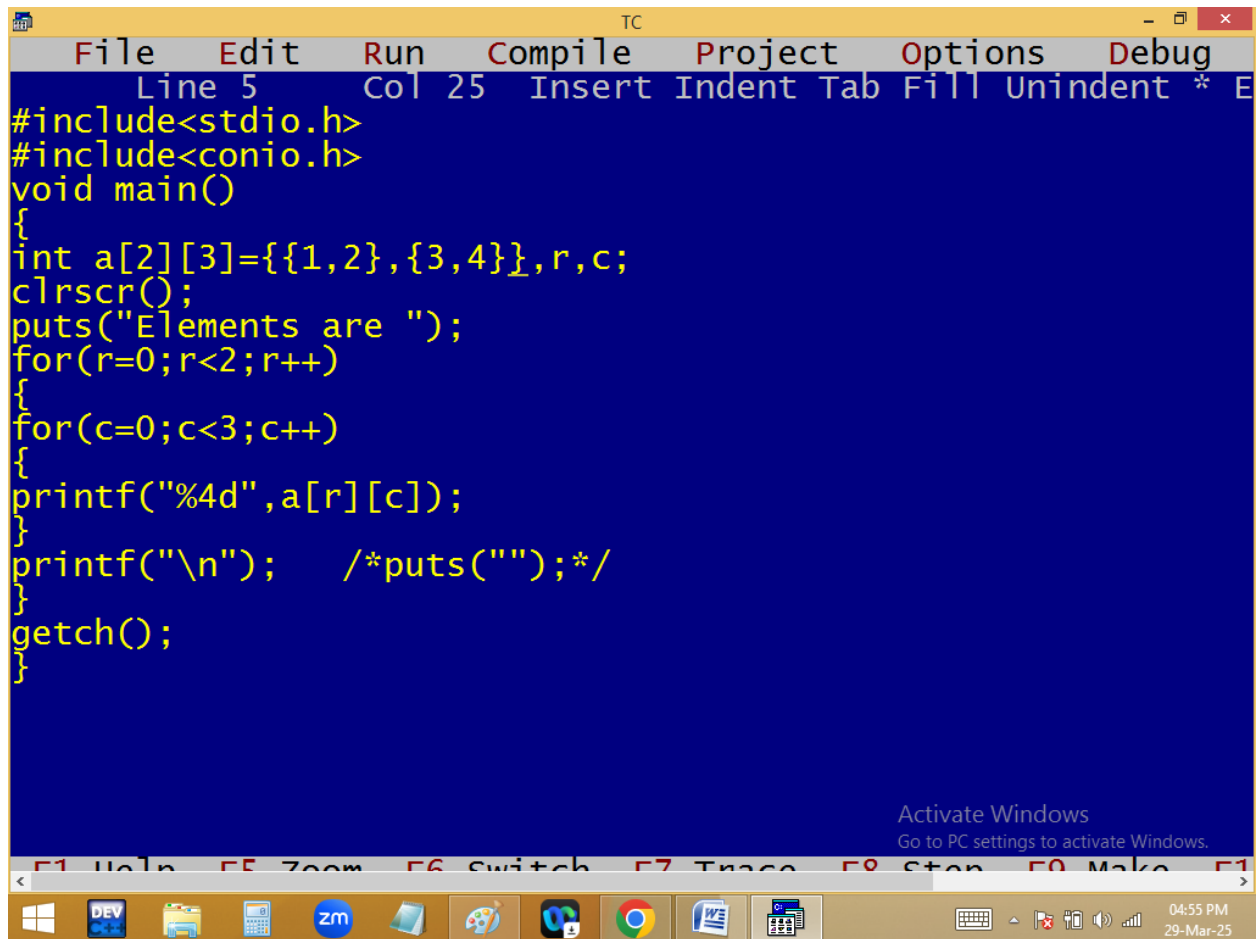


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 6", "Col 21", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a list of keyboard shortcuts: "F1", "F2", "F3", "F4", "F5", "F6", "F7", "F8", "F9", "F10", "F11", "F12". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3},4,5,6},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, a folder, a calculator, a Zoom icon, a paint palette, a code editor, a Chrome browser, a Word document, and a calculator. On the right side of the taskbar, the system clock shows "04:53 PM" and the date "29-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

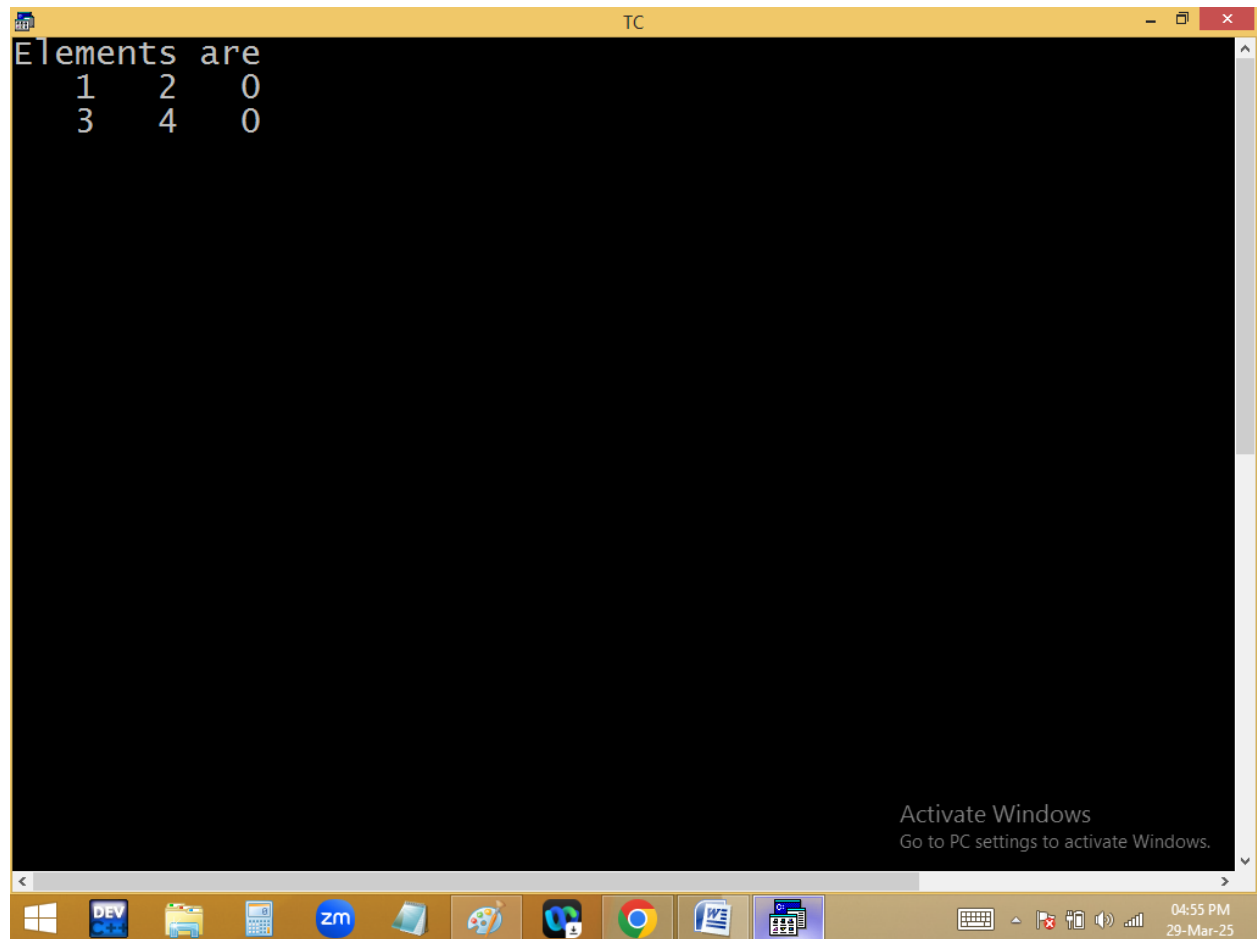


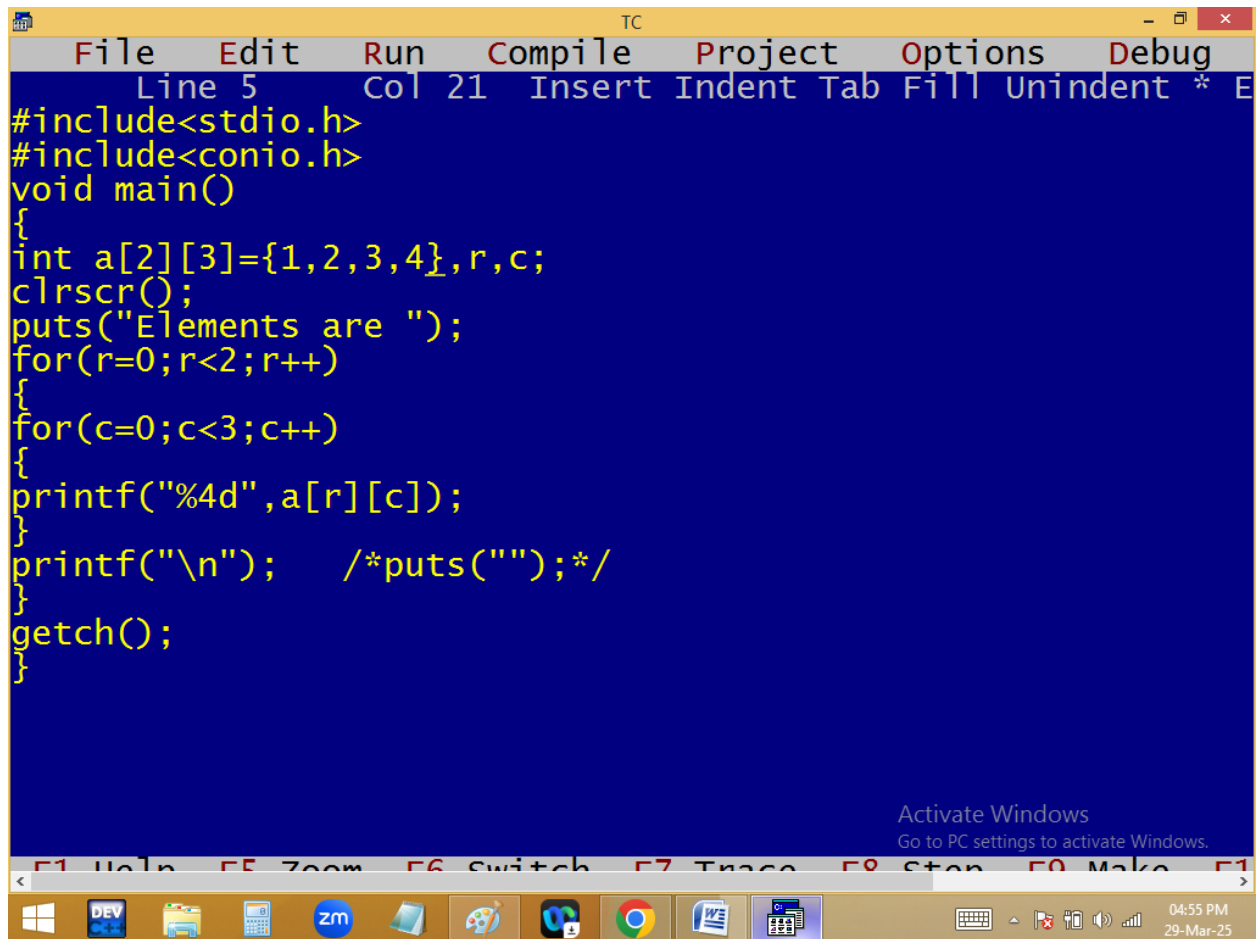


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 5", "Col 25", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2},{3,4}},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a taskbar with icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. A system tray on the right shows the time "04:55 PM" and date "29-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

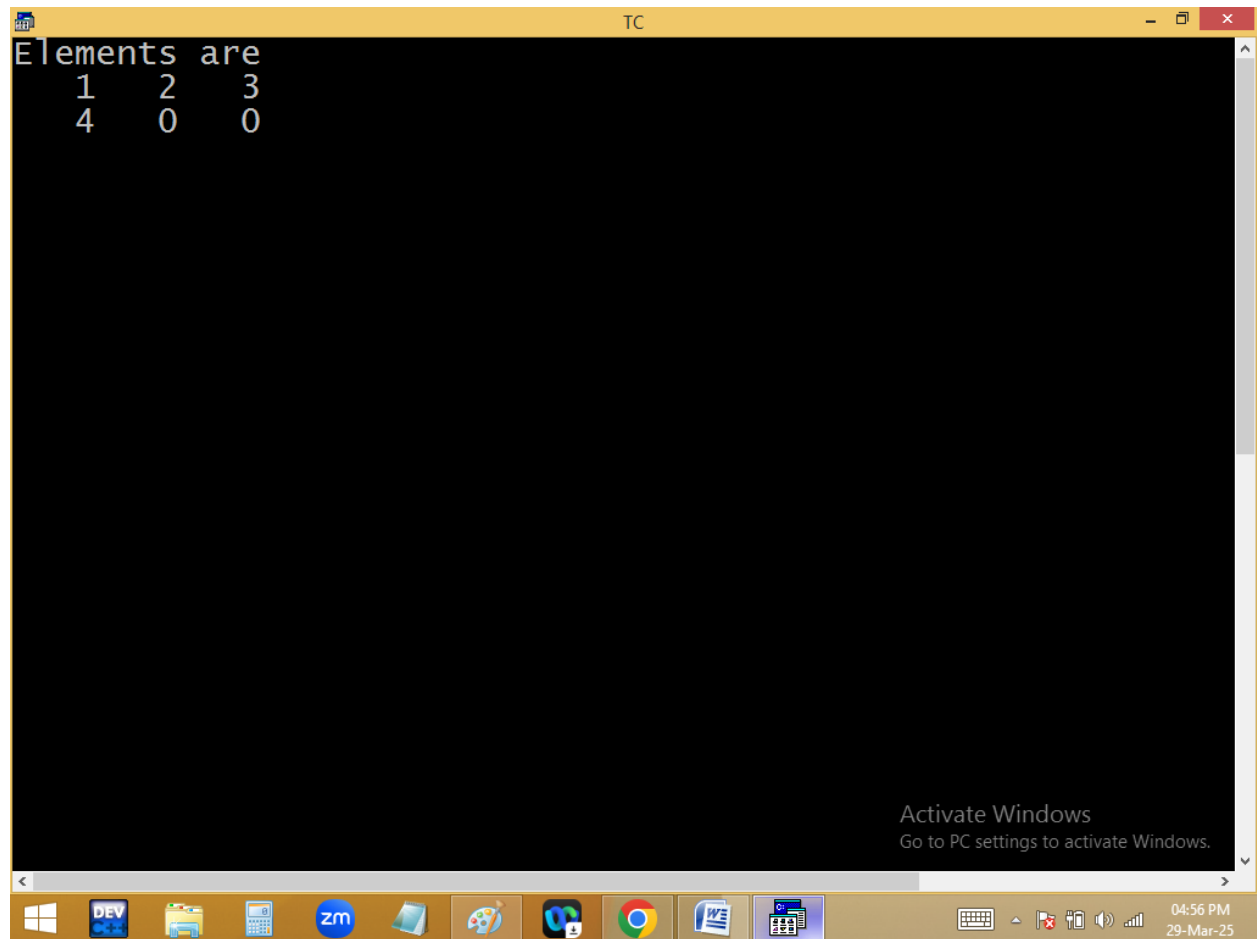


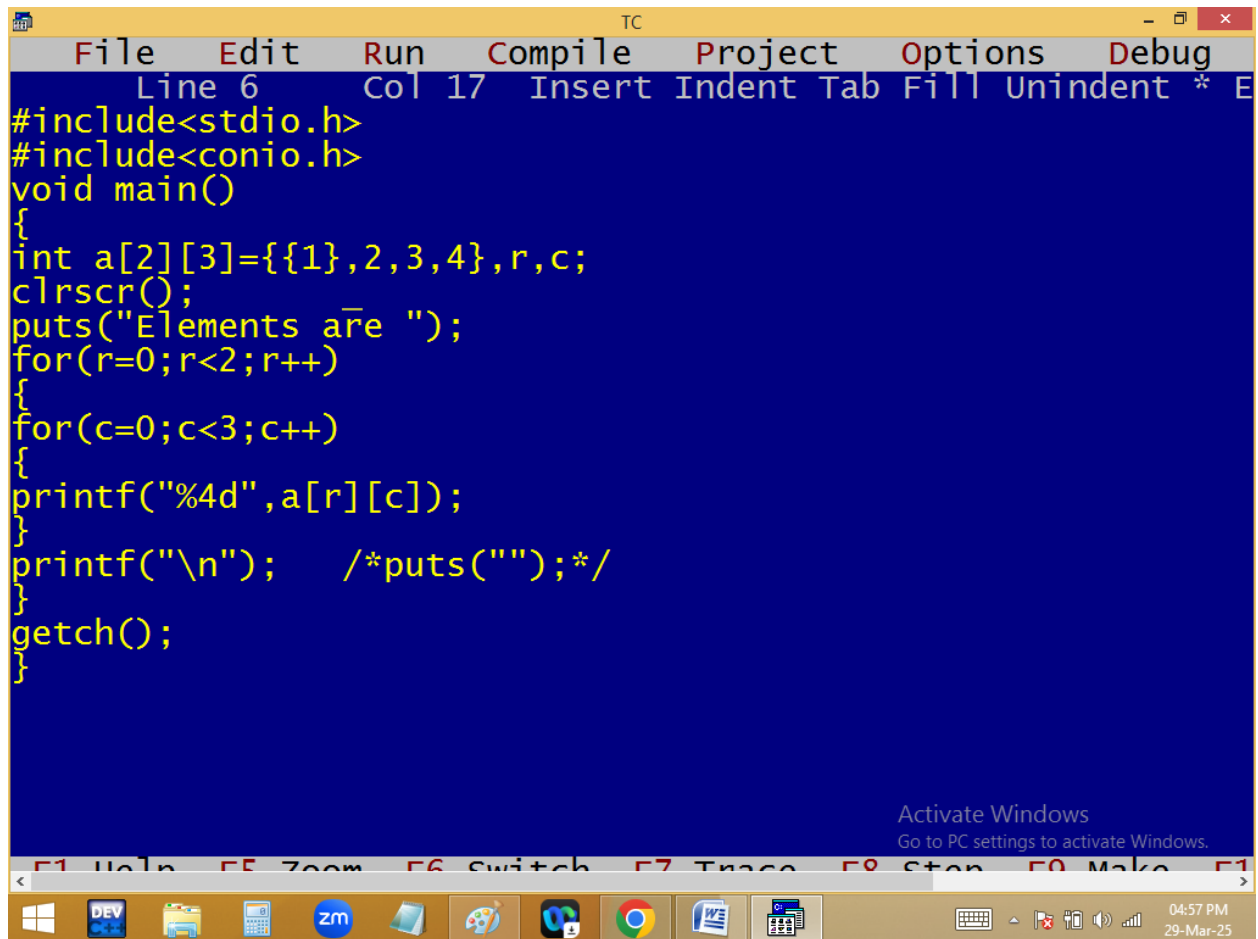


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 5" and "Col 21". The main editing area has a blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time: "04:55 PM 29-Mar-25".

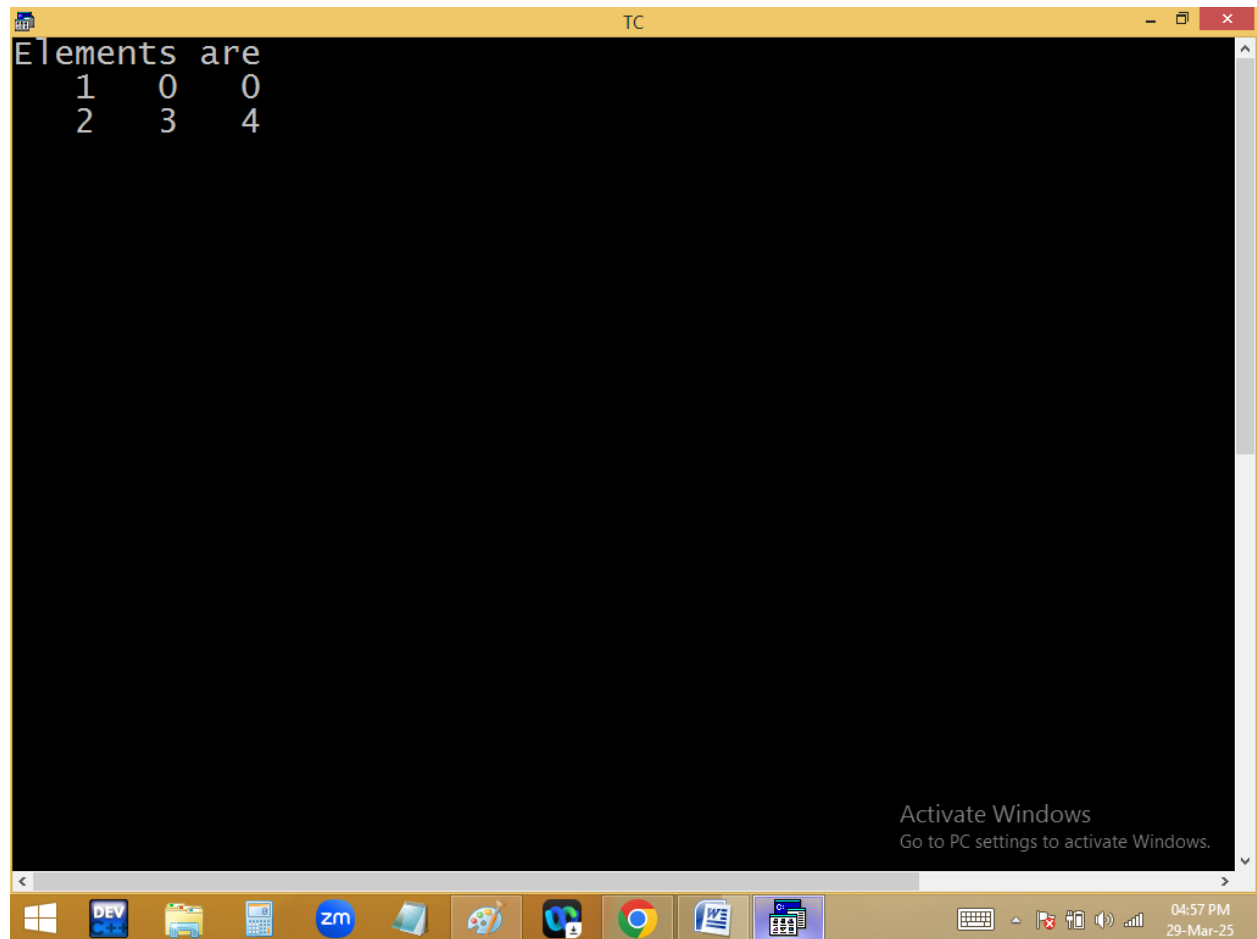




The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 6", "Col 17", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1},2,3,4},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom right of the IDE window, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system clock in the bottom right corner shows "04:57 PM" and "29-Mar-25".



TC

File Edit Run Compile Project Options Debug

Error: Too many initializers in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1},{2},3,4},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

04:58 PM
29-Mar-25

TC

File Edit Run Compile Project Options Debug

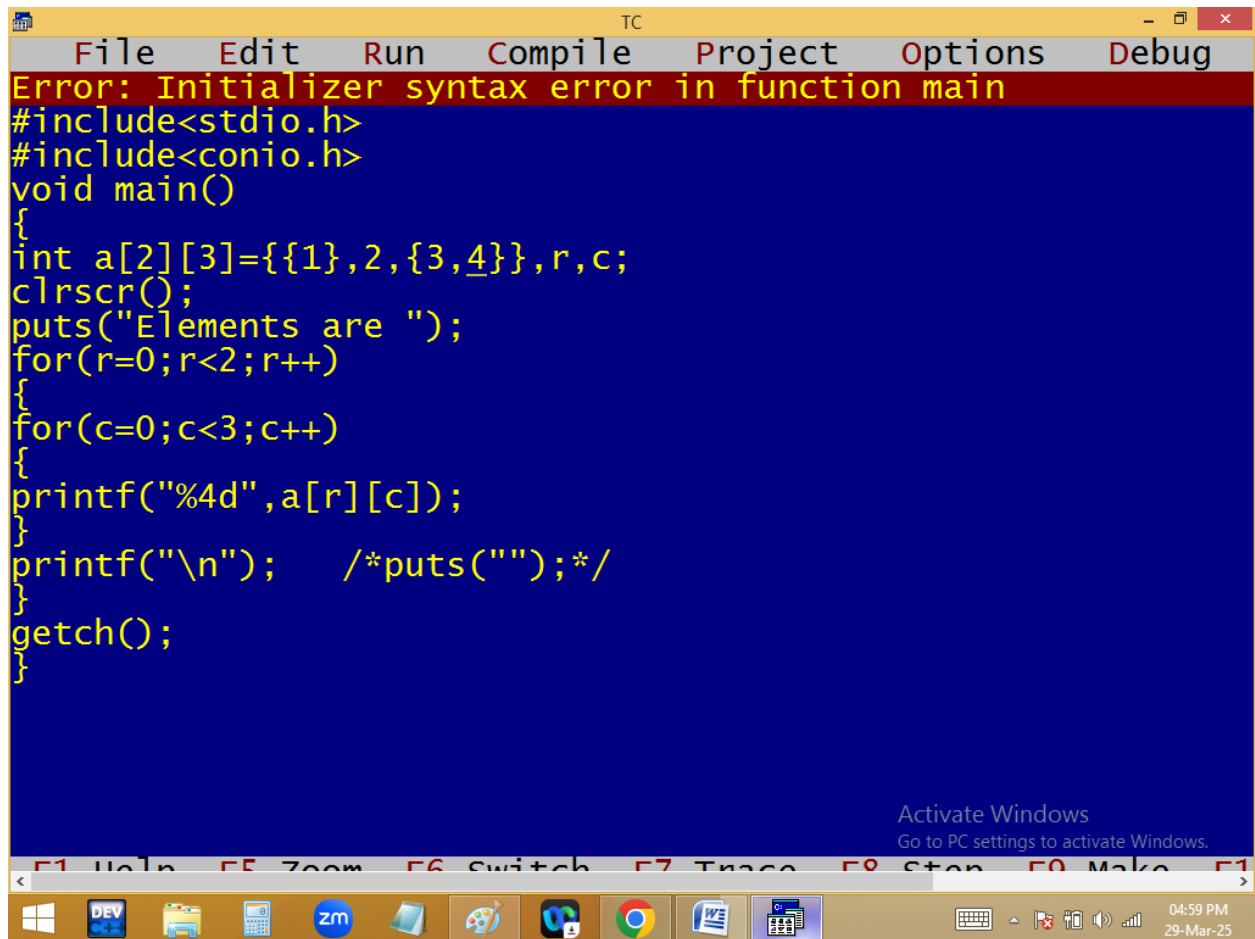
Error: Too many initializers in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2},3,4,5,6,r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

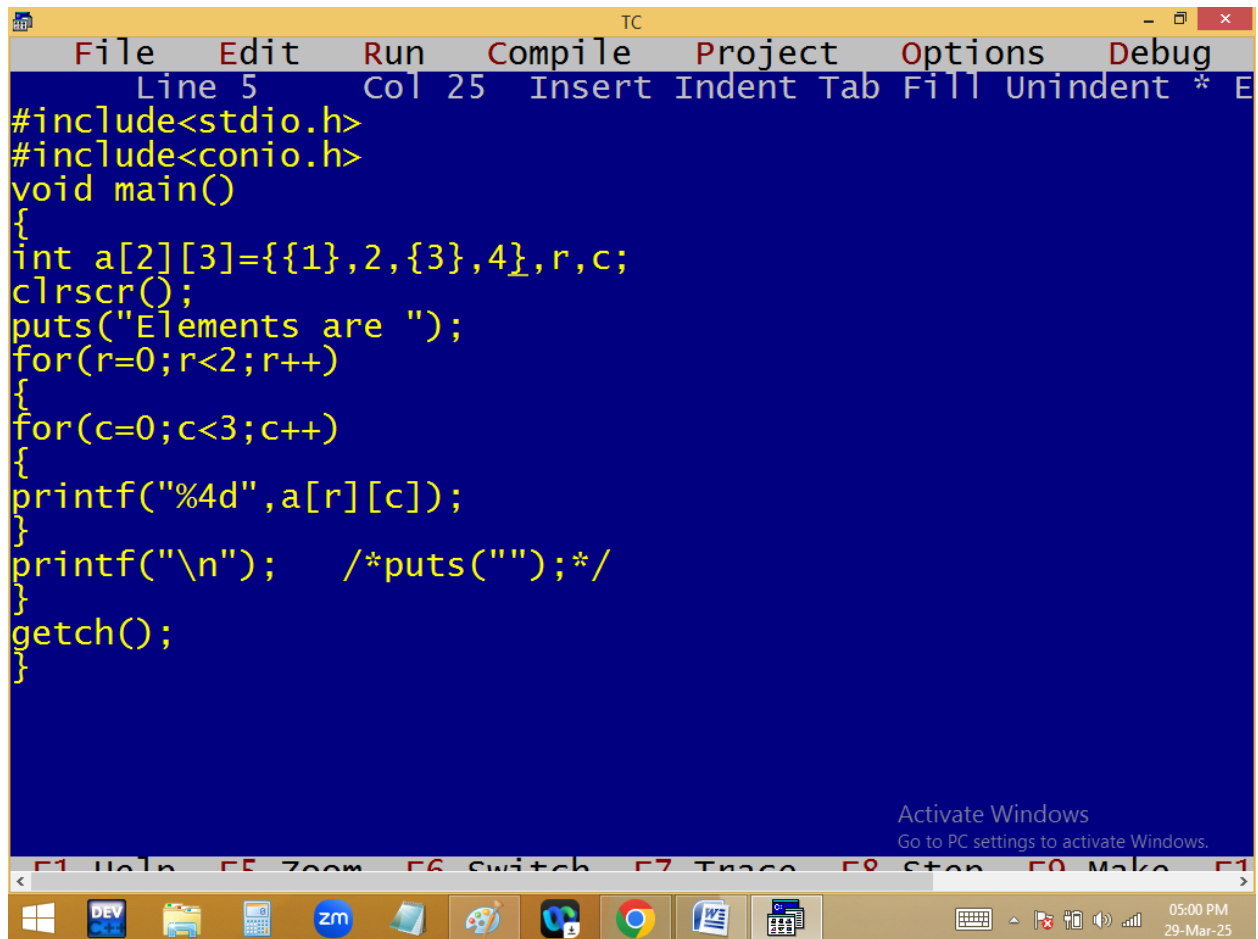
04:53 PM
29-Mar-25



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC" and includes standard window controls. The menu bar contains "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code editor displays the message: "Error: Initializer syntax error in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1},2,{3,4}},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

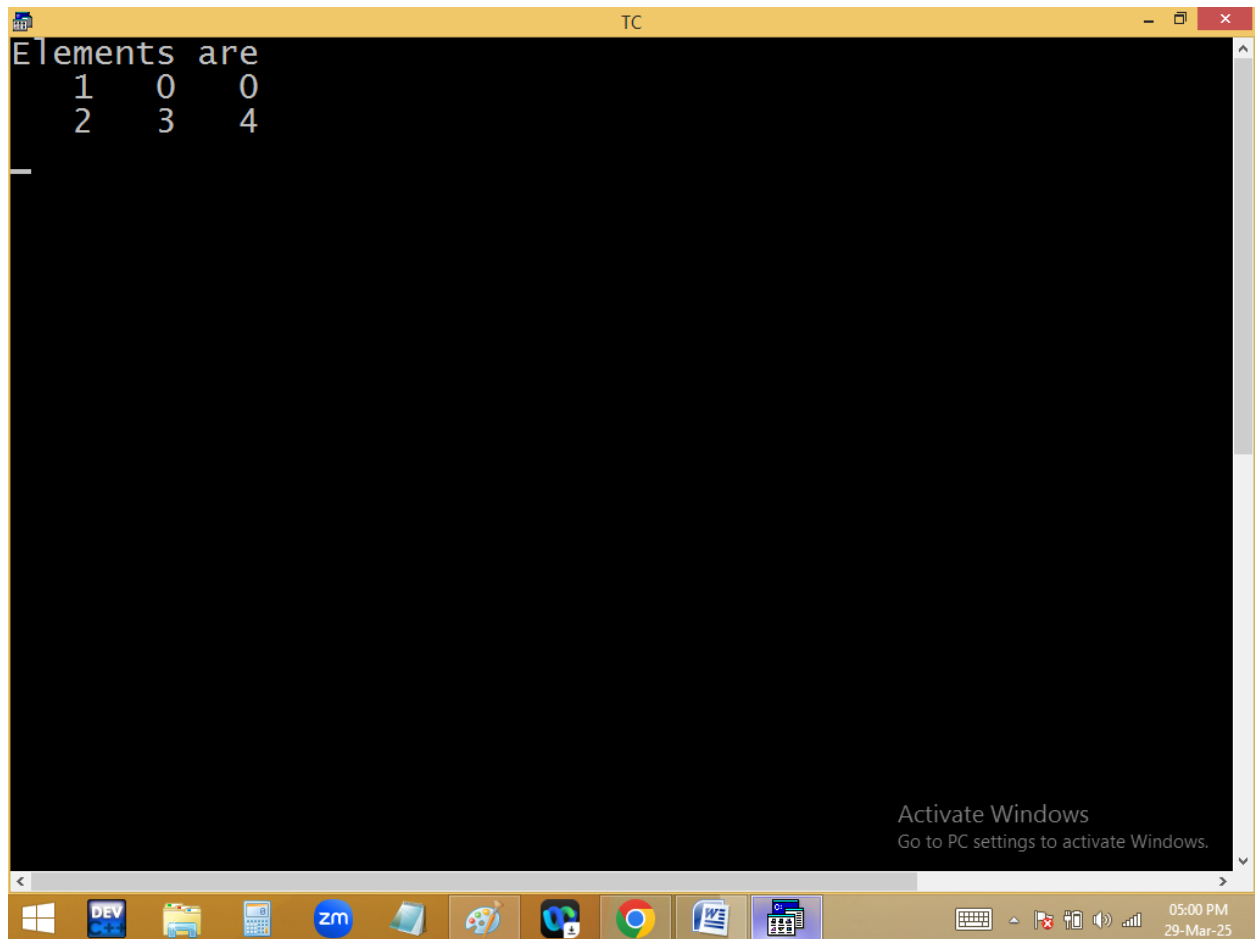
At the bottom of the window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the very bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner indicates the time is 04:59 PM on 29-Mar-25.

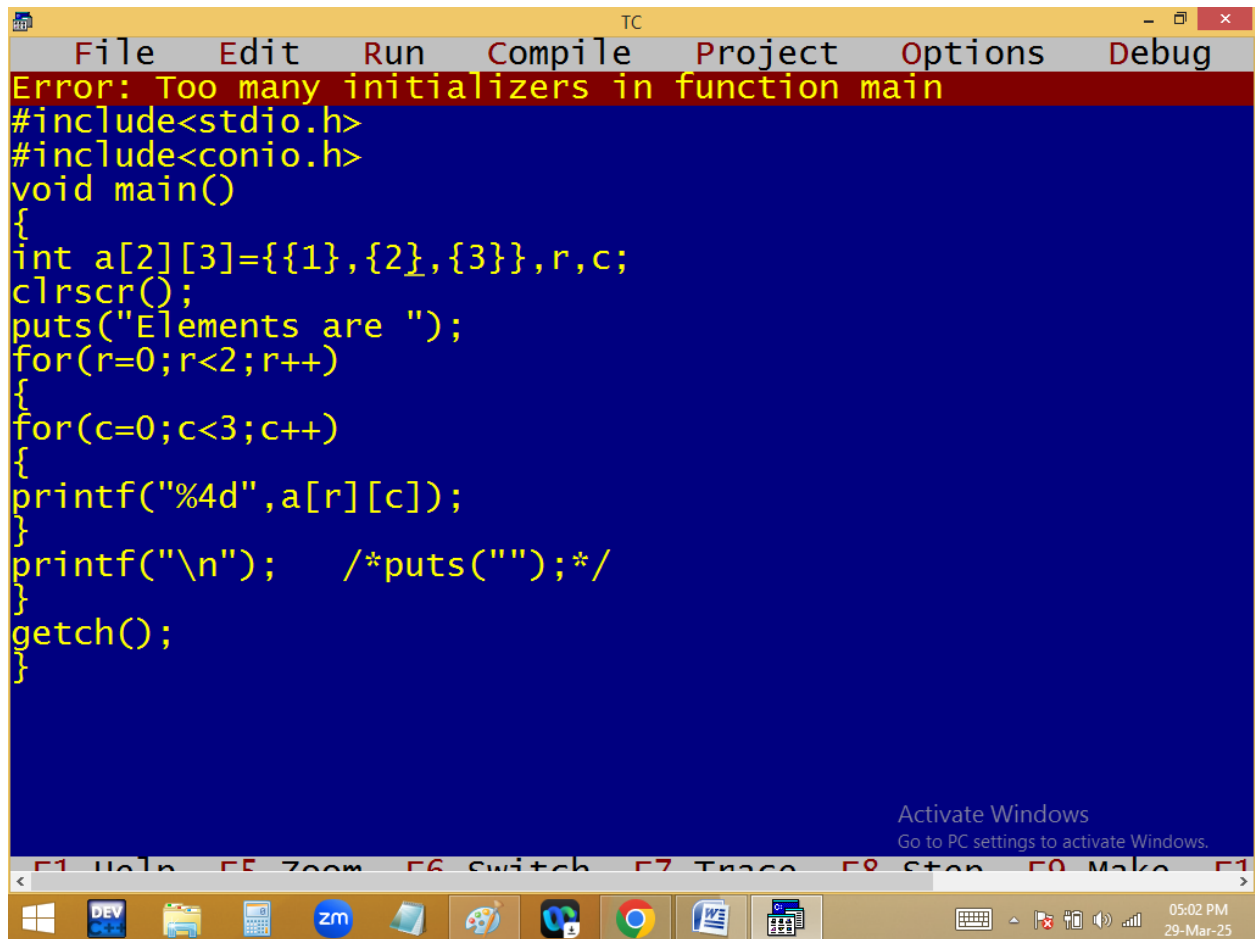


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 5", "Col 25", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a list of symbols: "*, E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1},2,{3},4},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time: "05:00 PM 29-Mar-25".





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the code area states "Error: Too many initializers in function main". The code being compiled is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1},{2},{3}},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. The Windows taskbar is visible at the very bottom, showing icons for Windows, DEV, File Explorer, Calculator, Zoom, and other applications. The system clock in the bottom right corner indicates the time is 05:02 PM on 29-Mar-25.

TC

File Edit Run Compile Project Options Debug

Error: Size of structure or array not known in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[][]={1,2,3,4,5,6},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:03 PM
29-Mar-25

TC

File Edit Run Compile Project Options Debug

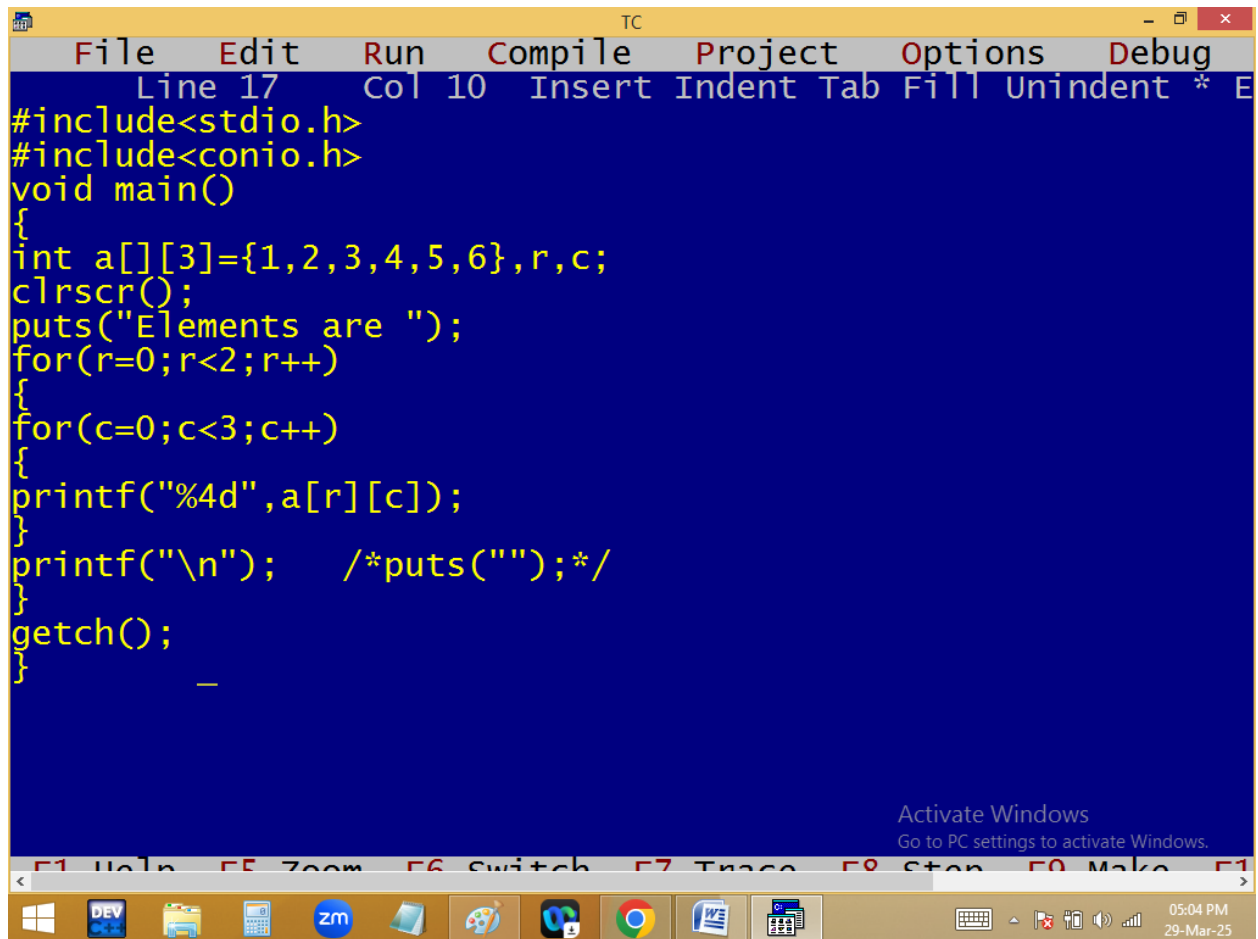
Error: Size of structure or array not known in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][]={1,2,3,4,5,6},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

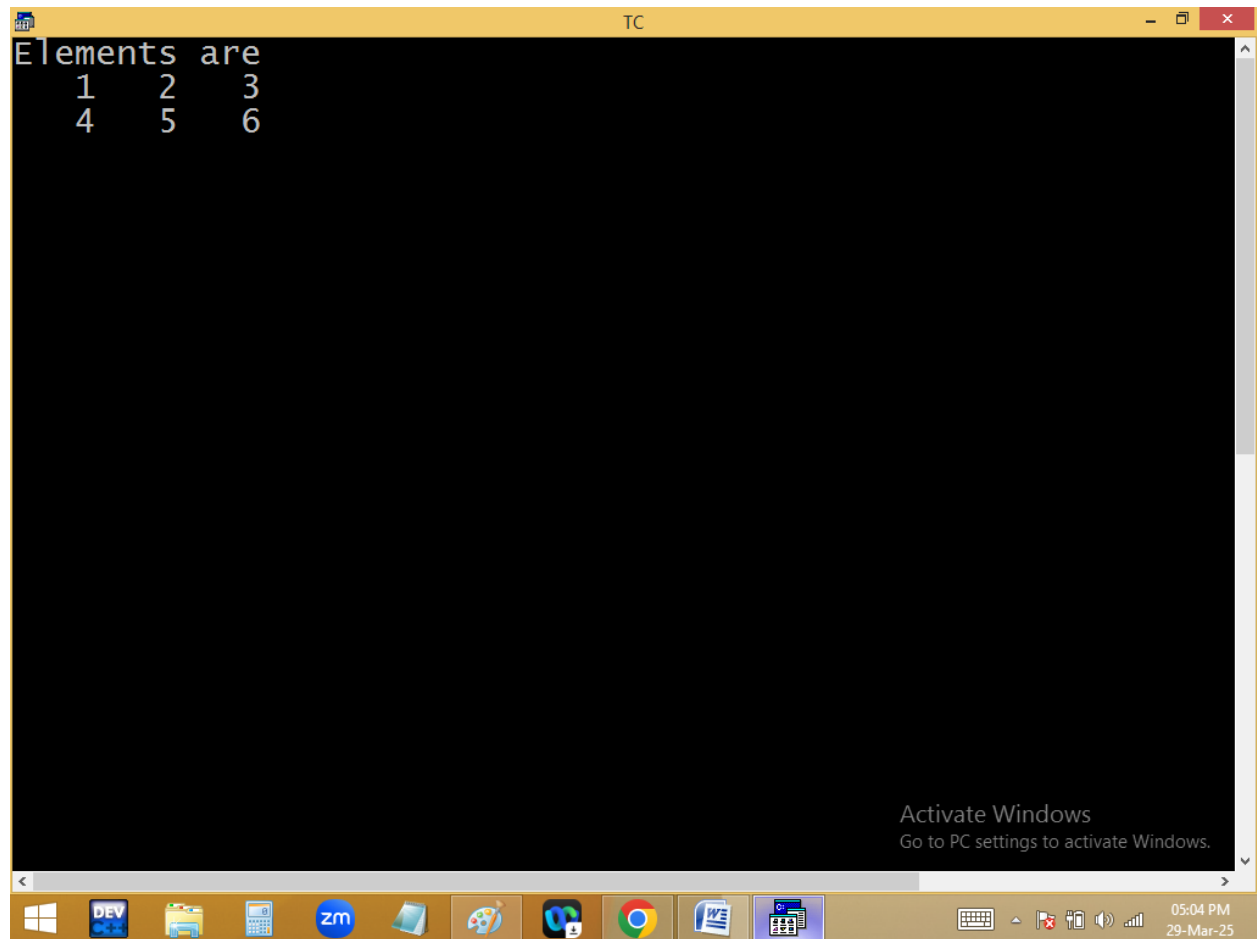
05:03 PM
29-Mar-25

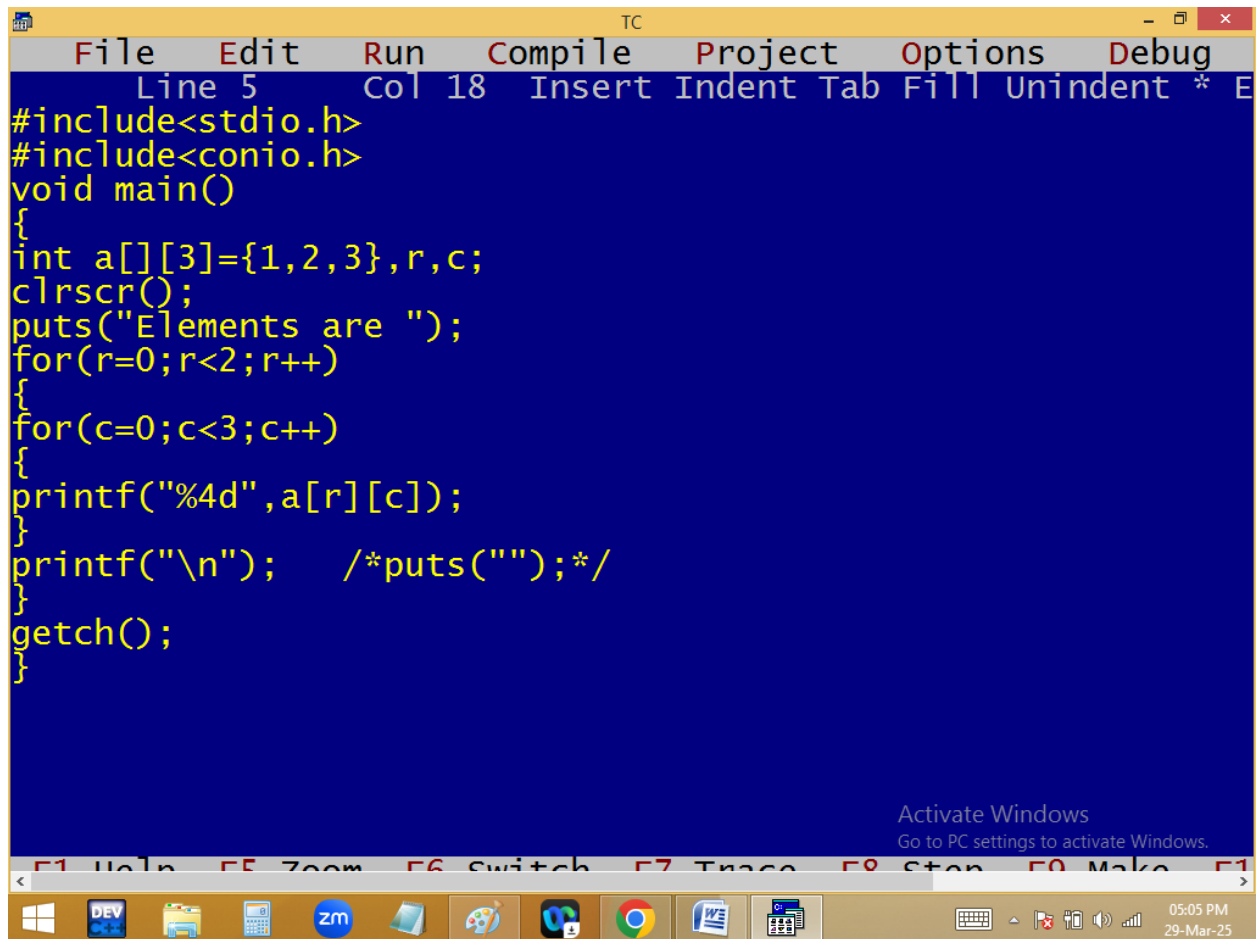


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 17", "Col 10", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[][3]={1,2,3,4,5,6},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the IDE window, there is a text prompt: "Activate Windows. Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, Paint, VS Code, Chrome, Word, and Excel. The system tray on the right shows the date and time: "05:04 PM 29-Mar-25".

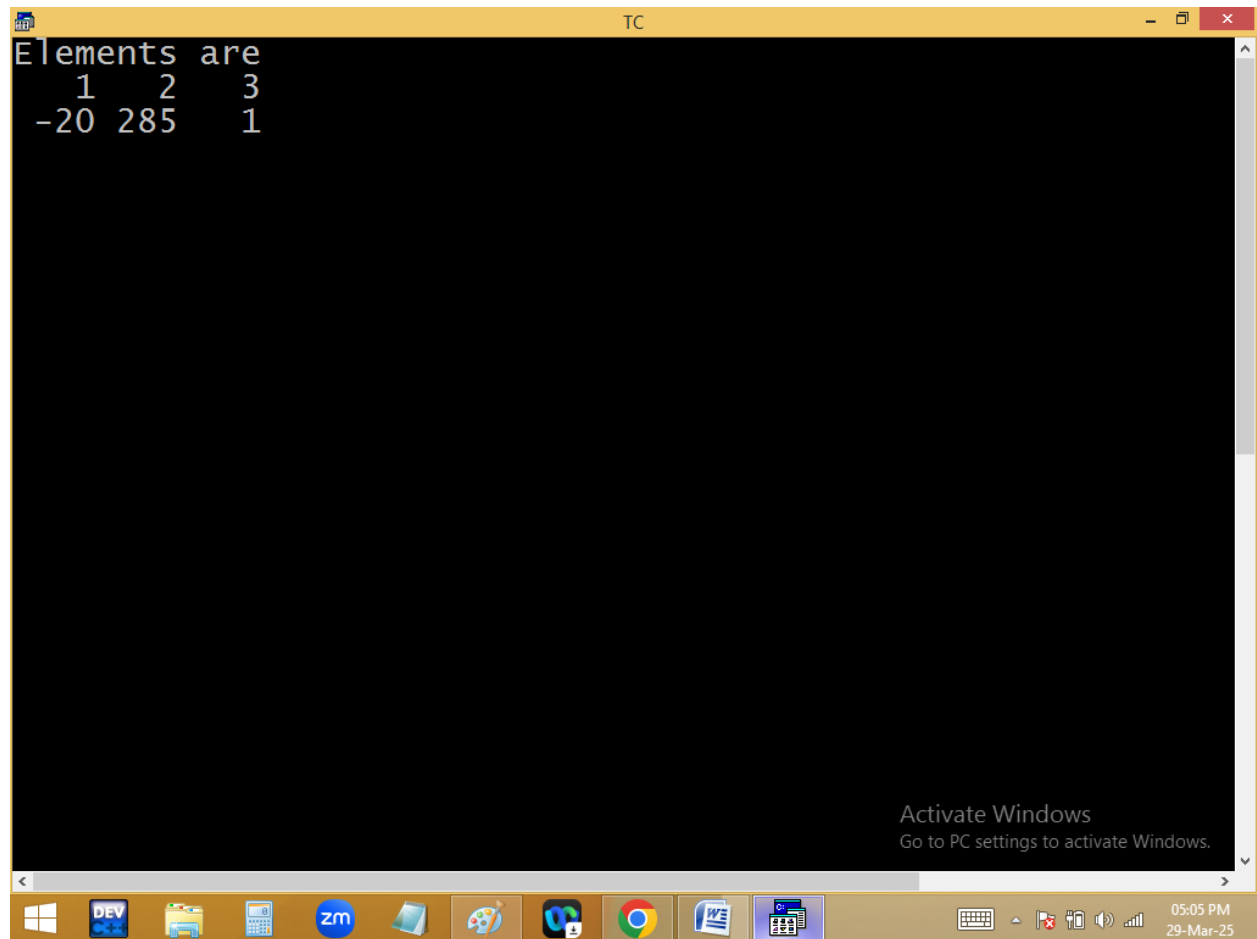


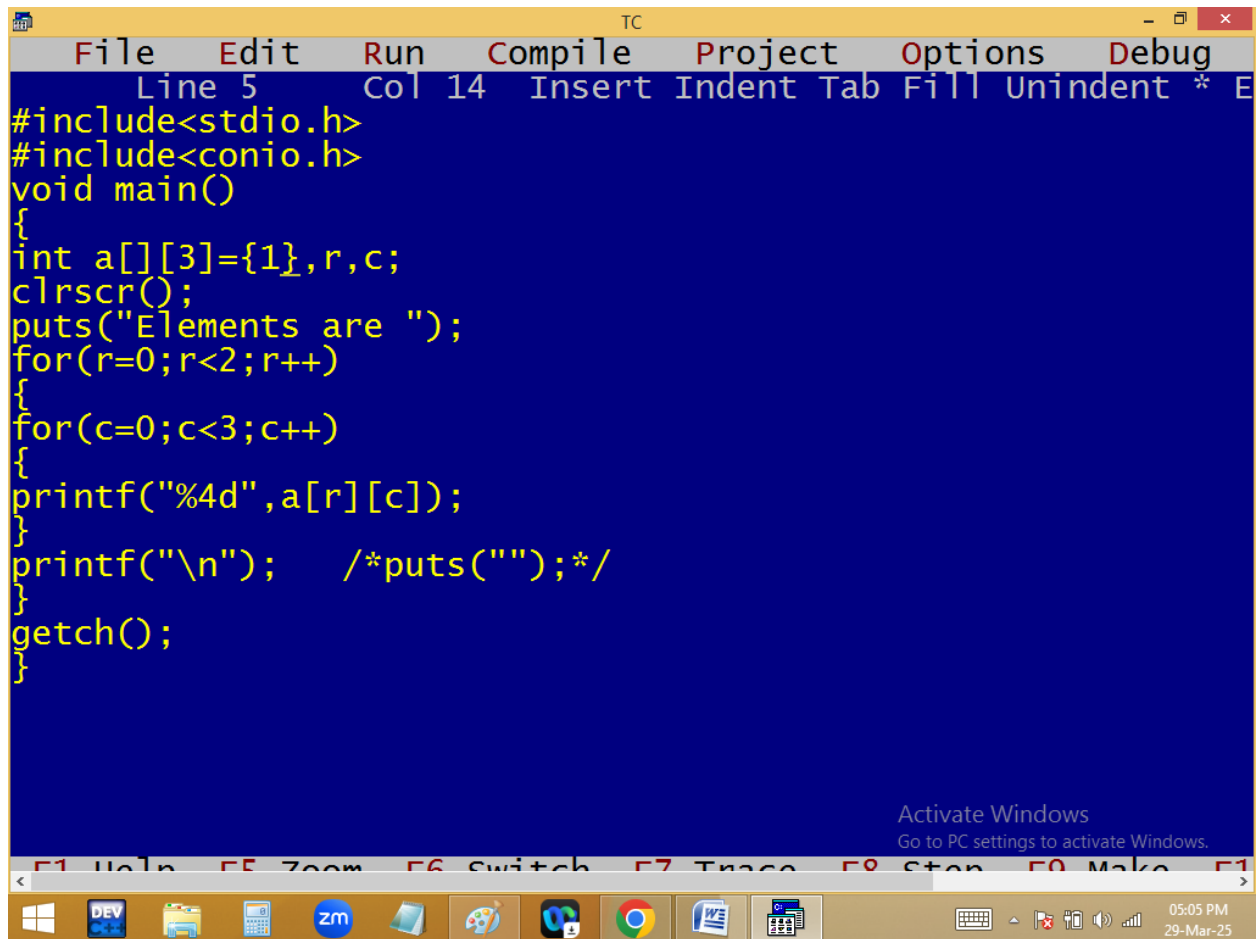


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status line indicates "Line 5" and "Col 18", followed by menu options: "Insert", "Indent", "Tab", "Fill", "Unindent", and a list of keyboard shortcuts: "*, E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[][3]={1,2,3},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, a folder, a calculator, a Zoom icon, a paint palette, a code editor, Google Chrome, a document, and a calculator. On the right side of the taskbar, the system clock shows "05:05 PM" and the date "29-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, with the text "Go to PC settings to activate Windows."





The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 5" and "Col 14". The main editing area has a blue background with yellow text. The code is as follows:

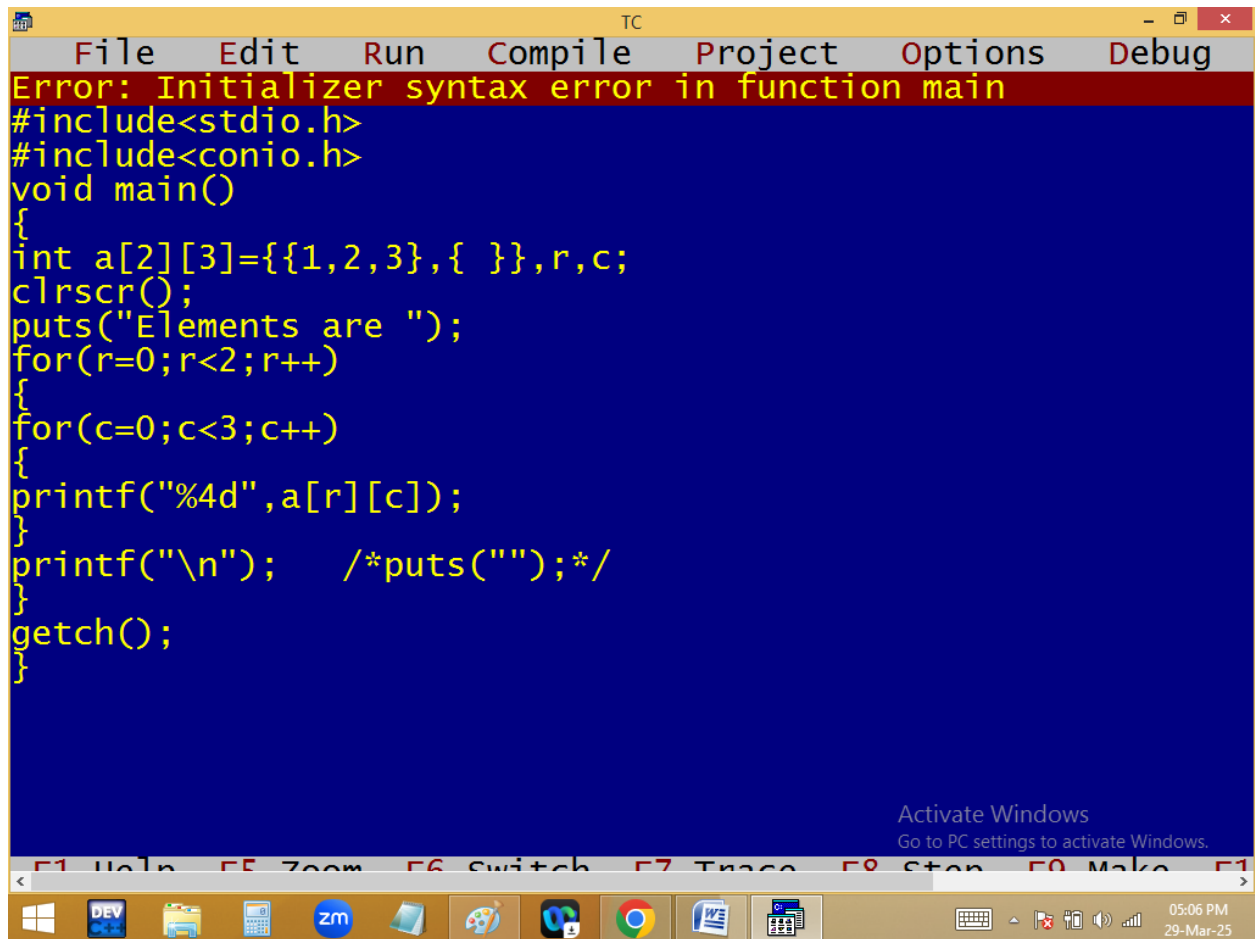
```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[][3]={1},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows, DEV, a folder, a calculator, a Zoom icon, a paint palette, a code editor icon, a Chrome browser icon, a Word document icon, and a calculator icon. On the right side of the taskbar, the system clock shows "05:05 PM" and the date "29-Mar-25". A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.

```
TC
Elements are
  1  0  0
-20 285 1
```

Activate Windows
Go to PC settings to activate Windows.

05:06 PM
29-Mar-25



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC" and includes standard window controls. The menu bar contains "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". A red error banner at the top of the editor area states "Error: Initializer syntax error in function main". The code in the editor is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={{1,2,3},{ }},r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

At the bottom of the IDE window, there is a status bar with function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Stop, F9 Make, and F10. Below the IDE window is the Windows taskbar, which includes icons for Windows, DEV, File Explorer, Calculator, Zoom, and other applications. The system tray on the right shows the date and time as "05:06 PM 29-Mar-25".

TC

File Edit Run Compile Project Options Debug

Error: Incompatible type conversion in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]=9,r,c;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:07 PM
29-Mar-25

TC

File Edit Run Compile Project Options Debug

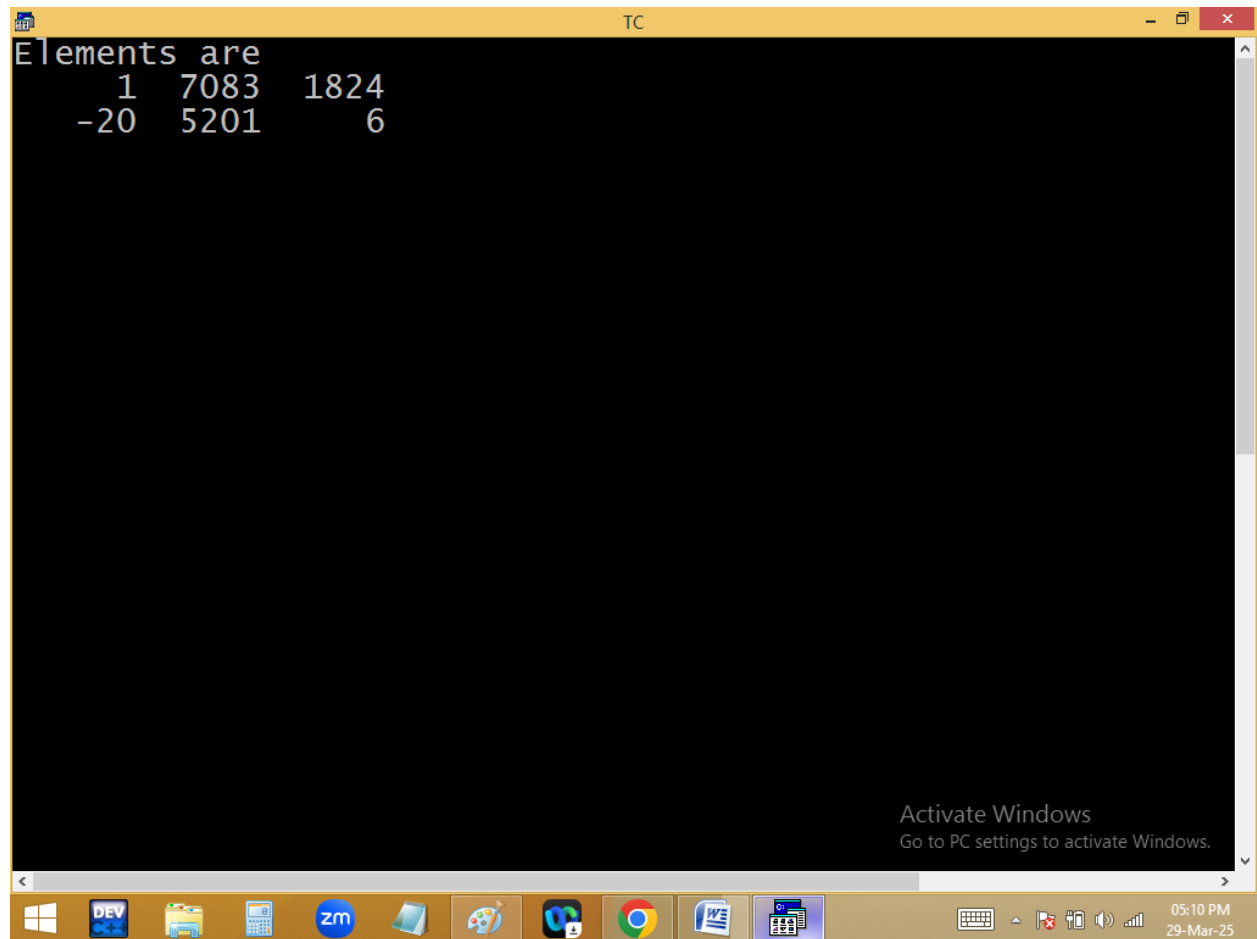
Line 13 Col 11 Insert Indent Tab Fill Unindent * E

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3],r,c;
a[0][0]=1;a[1][2]=6;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%6d",a[r][c]);
}
printf("\n");    /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:10 PM
29-Mar-25

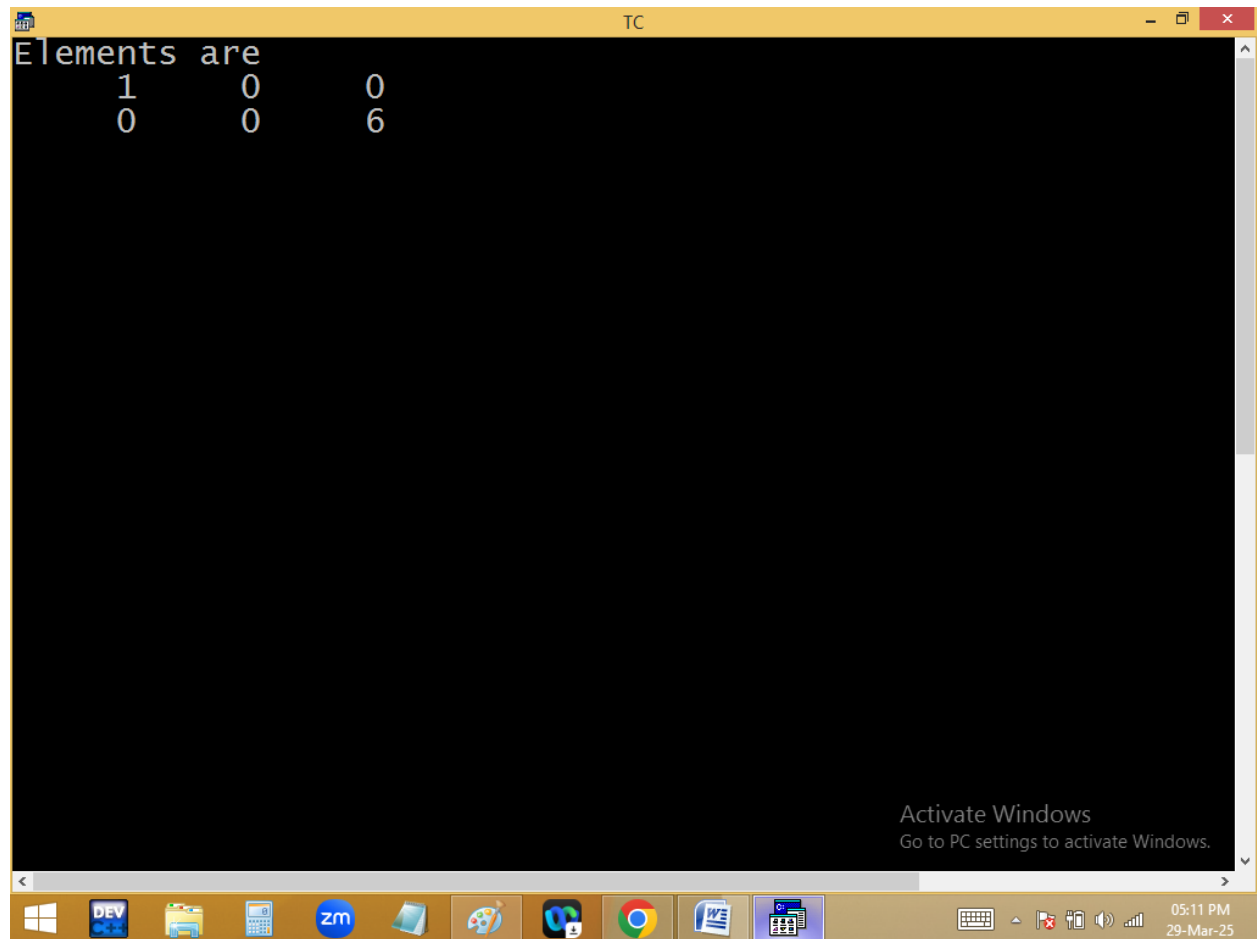


```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 17 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={-1},r,c;
a[0][0]=1;a[1][2]=6;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%6d",a[r][c]);
}
printf("\n"); /*puts("");*/
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

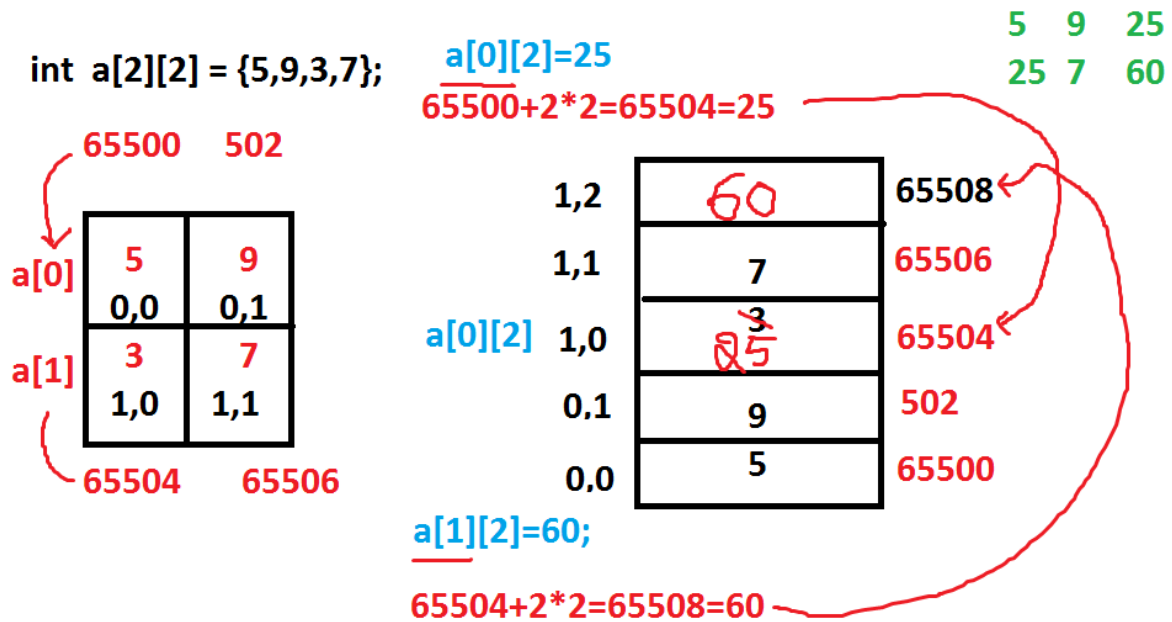
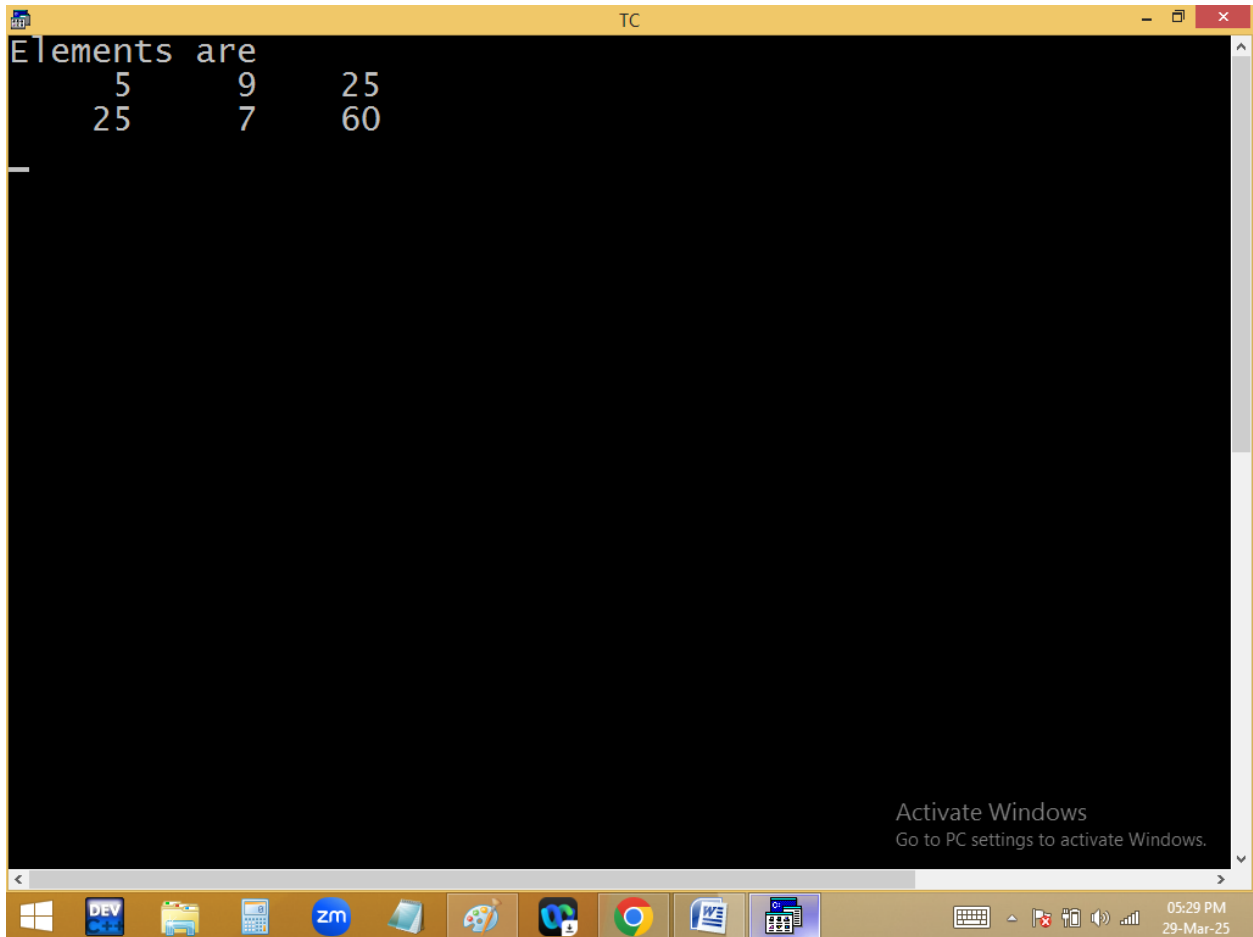
05:11 PM
29-Mar-25



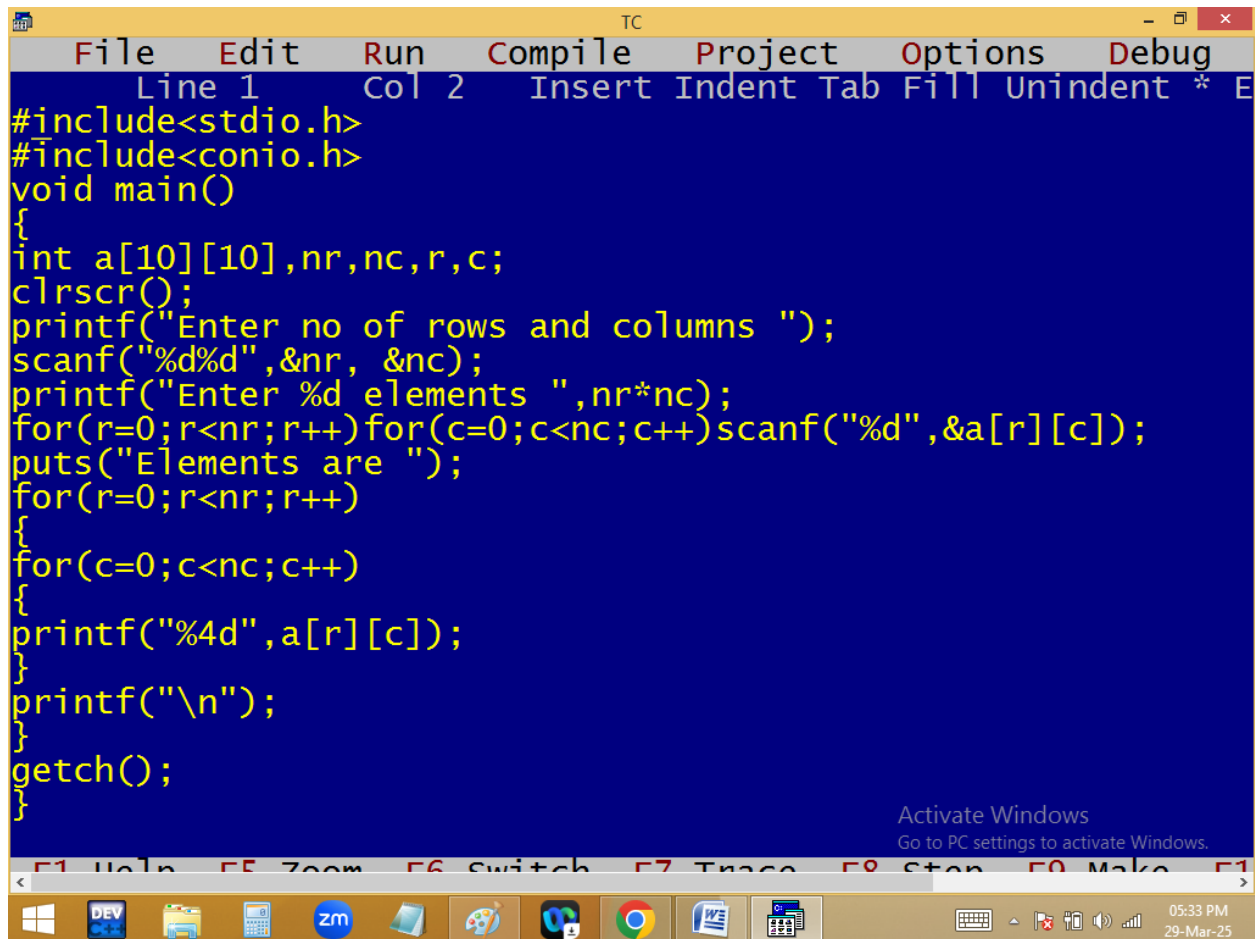

```
TC
File Edit Run Compile Project Options Debug
Line 21 Col 11 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][2]={5,9,3,7},r,c;
a[0][2]=25;a[1][2]=60;
clrscr();
puts("Elements are ");
for(r=0;r<2;r++)
{
for(c=0;c<3;c++)
{
printf("%6d",a[r][c]);
}
printf("\n"); /*puts("");*/
}
getch();
}
/*
5 9 25
25 7 60 */_

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10
```



Reading and printing elements of m*n matrix:



The screenshot shows a Turbo C++ (TC) IDE window with a blue background. The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the bottom shows function key shortcuts: F1 Help, F5 Zoom, F6 Switch, F7 Trace, F8 Step, F9 Make, and F10. The code is written in yellow text on the blue background. It includes headers for stdio.h and conio.h, defines a main function, declares a 10x10 integer array 'a', and uses clrscr() to clear the screen. It prompts the user to enter the number of rows and columns, reads these values, and then prompts for the matrix elements. The matrix is read using a nested loop and printed back using another nested loop with a tab separator and a newline character. The program ends with getch() to keep the window open.

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc,r,c;
clrscr();
printf("Enter no of rows and columns ");
scanf("%d%d",&nr, &nc);
printf("Enter %d elements ",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("Elements are ");
for(r=0;r<nr;r++)
{
for(c=0;c<nc;c++)
{
printf("%4d",a[r][c]);
}
printf("\n");
}
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

05:33 PM
29-Mar-25

```
TC
Enter no of rows and columns 2 3
Enter 6 elements 2 4 0 1 9 3
Elements are
    2    4    0
    1    9    3
```

Activate Windows
Go to PC settings to activate Windows.

05:33 PM
29-Mar-25

```
TC
Enter no of rows and columns 3 3
Enter 9 elements 1 2 0 3 4 7 8 4 6
Elements are
  1  2  0
  3  4  7
  8  4  6
```

Activate Windows
Go to PC settings to activate Windows.

Finding array element , position and address of $n \times n$ matrix:

The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 9", "Col 1", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
for(r=0;r<2;r++)
for(c=0;c<3;c++)
printf("a[%d][%d]value=%d, addr=%u\n",r,c,a[r][c],&a[r][c]);
getch();
}
```

At the bottom of the window, there is a taskbar with several icons: Windows Start button, DEV, File Explorer, Calculator, Zoom, a folder icon, Paint, VS Code, Google Chrome, Word, and a calculator. To the right of the taskbar, a system tray shows the date and time: "05:37 PM 29-Mar-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.

```
TC
a[0][0]value=1, addr=65492
a[0][1]value=2, addr=65494
a[0][2]value=3, addr=65496
a[1][0]value=4, addr=65498
a[1][1]value=5, addr=65500
a[1][2]value=6, addr=65502
```

Activate Windows
Go to PC settings to activate Windows.

05:37 PM
29-Mar-25

Online compiler:

C Language by Kishore Sir Java B x Online C Compiler - Programiz x

programiz.com/c-programming/online-compiler/WhatsApp Classroom Gmail Proctur - Simplifyin... YouTube movie rulez2 com 2... Movierulz | Watch B... 22 wrapper

Programiz

C Online Compiler

Programiz PRO >

main.c

Run

```
1 #include<stdio.h>
2
3 int main()
4 {
5     int a[2][3]={1,2,3,4,5,6},r,c;
6
7     for(r=0;r<2;r++)
8     for(c=0;c<3;c++)
9     printf("a[%d][%d]value=%d, addr=%u\n",r,c,a[r][c]
10           ,&a[r][c]);
11 }
```

Output

Clear

a[0][0]value=1, addr=1884636160
a[0][1]value=2, addr=1884636164
a[0][2]value=3, addr=1884636168
a[1][0]value=4, addr=1884636172
a[1][1]value=5, addr=1884636176
a[1][2]value=6, addr=1884636180

=== Code Execution Successful ===

Activate Windows
Go to PC settings to activate Windows.

Connecting...

DEV

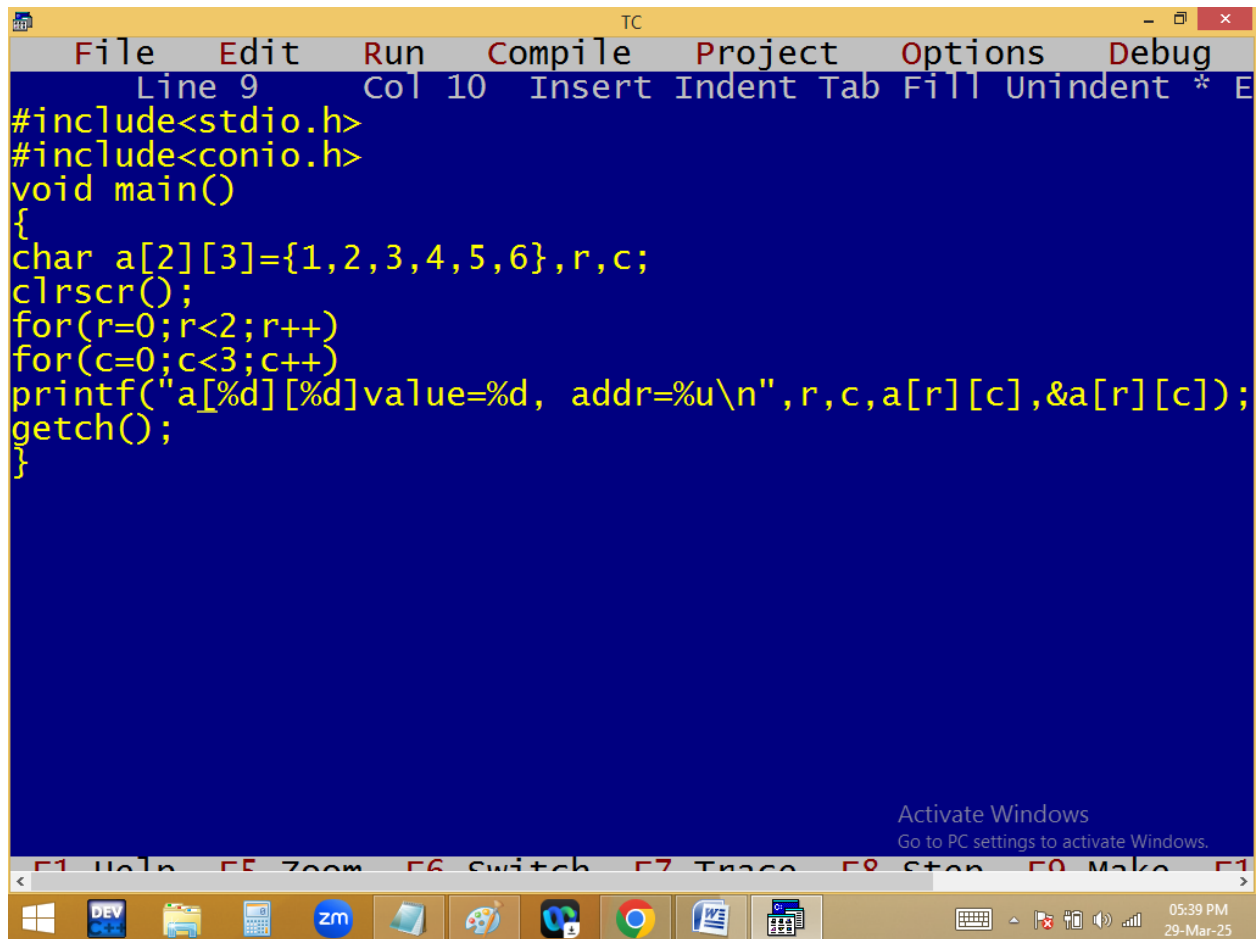
zom

Chrome

VS Code

Calculator

05:39 PM 29-Mar-25



```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 10 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
char a[2][3]={1,2,3,4,5,6},r,c;
clrscr();
for(r=0;r<2;r++)
for(c=0;c<3;c++)
printf("a[%d][%d]value=%d, addr=%u\n",r,c,a[r][c],&a[r][c]);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

05:39 PM
29-Mar-25

```
TC
a[0][0]value=1, addr=65496
a[0][1]value=2, addr=65497
a[0][2]value=3, addr=65498
a[1][0]value=4, addr=65499
a[1][1]value=5, addr=65500
a[1][2]value=6, addr=65501
```

Activate Windows
Go to PC settings to activate Windows.

Home work:

1. Transpose of $n \times n$ matrix:

2*3

1	2	3
4	5	6

3*2

1	4
2	5
3	6

2. Find the trace of $n \times n$ matrix.

1	2	3
4	5	6
0	9	7

trace=13

3. Trace of right diagonal elements.

1	2	3
4	5	6
0	9	7

sum=8